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Intellectual Property Challenges in Corporate-Backed Ventures

Development of a Global Intellectual Property Playbook for Scania CV AB

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Sammanfattning

Flera immaterialrättsliga utmaningar kan uppstå inom samarbetet mellan olika parter. Samarbetsformen är en faktor som påverkar de immaterialrättsliga utmaningar som behöver hanteras. Till exempel kan samarbetet innehålla gemensam forskning och utveckling inom ett projekt. Uppstartsbolag, universitet och företagsriskkapitalbolag, det vill säga den investeringsarm hos etablerade företag som investerar i uppstartsbolag, är exempel på parter som kan vara involverade i samarbetet. Att anpassa operativa och strategiska mål för involverade parter, genomföra kontraktsförhandlingar och säkra immaterialrättsliga intressen kan vara utmanande. Därför är det praktiskt att ha ett verktyg som underlättar hanteringen av detta. Ett exempel på ett sådant verktyg är en “playbook” som tillhandahåller praktiska riktlinjer och är enkel att använda. Syftet med detta examensarbete, vilket baseras på en fallstudie genomförd i samarbete med Scania CV AB, var att utreda immaterialrättsliga nyckelutmaningar i “corporate-backed ventures” och ta fram praktiska riktlinjer för hantering av dessa. En “Global Intellectual Property Playbook” togs fram, i vilken de praktiska riktlinjerna sammanställdes, med syfte att skapa transparens i de immaterialrättsliga processerna inom samarbetet mellan ett etablerat företag och ett uppstartsbolag. Detta möjliggör effektiv förhandling mellan parterna, vilket stärker samarbetet och de gemensamma forskningsaktiviteterna.

Nyckelord

- Etablerat företag: Ett företag som redan är etablerat på en marknad eller en industri. Lojal kundbas och lägre genomsnittliga kostnader än utmanande leverantörer eller varumärken är några exempel på möjliga fördelar för ett etablerat företag.[1]
- Uppstartsbolag: Ett nybildat företag med särskilt momentum som baseras på efterfrågan för den egna produkten eller tjänsten. Dessa kan växa snabbt som ett resultat av att erbjuda någonting som löser en viss marknadsklyfta.[2]
- Företagsriskkapital: Det handlar om att direktinvestera kapital i externa uppstartsbolag. Vanligtvis görs detta av etablerade företag som vill göra en liten investering i innovation. Högt förekommande är att etablerade företag stöttar uppstartsbolag med ledning, strategisk riktning och marknadsföringsexpertis.[3]
- Immaterialrätt: Avser uppfinningar, litterärt arbete, konstnärligt arbete, mönster, symboler, namn och bilder som används i handeln. Patent, upphovsrätter, varumärken, industridesigner och företagshemligheter är exempel på skydd av immateriella rättigheter.[4]
- Forskning och utveckling: Omfattar de aktiviteter som företagen behöver utföra för att åstadkomma innovation, introducera nya produkter och tjänster samt förbättra deras befintliga erbjudanden.[5]

Abstract

Several Intellectual Property (IP) challenges could arise in the cooperation between different parties. The cooperation form is one factor which influences the IP challenges that need to be managed. For example, the cooperation could include joint Research and Development (R&D) within a project. Start-ups, universities, and Corporate Venture Capital (CVC) firms, i.e. the investment arm of incumbent firms that invest in start-ups, are examples of parties that can be involved in the cooperation. Aligning operational and strategic goals of involved parties, conducting contract negotiations, and securing IP interests could be challenging. Therefore, it is practical to have a tool that facilitates the management of this. One example of such a tool is a playbook, which provides practical guidelines and is easy to use. The purpose of this thesis work, which is based on a case study conducted in collaboration with Scania CV AB, was to investigate key IP challenges in corporate-backed ventures and develop practical guidelines for management of these. A Global IP Playbook was developed, in which the practical guidelines were compiled, with the purpose of creating transparency in the IP processes within the cooperation between an incumbent firm and a start-up. This enables efficient negotiation between the parties, which strengthens the cooperation and joint research activities.

Keywords

- Incumbent firm: A business already established in a market or an industry. A loyal base of customers and lower average costs than challenger suppliers or brands are some examples of possible advantages for an incumbent firm.[1]
- Start-up: A newly formed business with particular momentum based on demand for its product or service. As a result of offering something that addresses a particular market gap, these could grow rapidly.[2]
- Corporate Venture Capital: This is about directly investing corporate funds into external start-ups. Usually, this is done by incumbent firms that want to make a small investment in innovation. Commonly, incumbent firms support start-ups with management, strategic direction, and marketing expertise.[3]
- Intellectual Property: Refers to inventions, literary work, artistic work, designs, symbols, names, and images used in commerce. Patents, copyrights, trademarks, industrial designs, and trade secrets are examples of intellectual property protection.[4]
- Research and Development: Covers the activities which companies need to undertake when innovating, introducing new products and services, and improving their existing offerings.[5]

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Acronyms

IP Intellectual Property

CVC Corporate Venture Capital

POV Point Of View

VC Venture Capital

EU European Union

U.S. United States

R&D Research and Development

FTO Freedom To Operate

WIPO World Intellectual Property Organisation

JDA Joint Development Agreement

NDA Non-Disclosure Agreement

M&A Mergers and Acquisitions

AI Artificial Intelligence

VNB Ventures and New Business

TT Technology Transfer

PoC Proof of Concept

Chapter 1

Introduction

Intellectual Property (IP) can be challenging in the cooperation between a start-up and an incumbent firm which has invested Corporate Venture Capital (CVC) in the start-up. Incumbent firms often have more rigid processes, which can be perceived as slow and hampering for the smaller firm. Asymmetry in information and resources between the parties involved in the cooperation can also be challenging. To handle this, it is important to clarify what each part contributes to the cooperation. This could, for instance, be related to financial strength, knowledge, test facilities, user data, and management. Moreover, the knowledge needs to be evaluated from both a technical and IP Point Of View (POV). The technical POV means the technical area in which the involved parties are cooperating, meanwhile, the IP POV covers general IP challenges such as *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*. In addition to information and resources, conflicts of interest could also be problematic. For instance, if the incumbent firm wants to keep the invention a secret and postpone the filing of a patent application, meanwhile, the smaller firm wants to present the invention to other incumbent firms in a venture round to obtain investments.

In the article "The Business of Intellectual Property: A Literature Review of IP Management Research" by M. Holgersson et al.[6], it is stated that strategic IP management is under-explored. This covers the complexity of strategic IP management in the cooperation between a start-up and an incumbent firm which has invested CVC in the start-up. Due to the under-explored state, there is a need to investigate challenges in IP management and develop guidelines on how to handle these in corporate venturing. Without practical guidelines for IP management, the cooperation will be hampered, and important opportunities for technological progress will be lost.

The purpose of this master thesis project was to investigate IP challenges in corporate-backed ventures and, based on this, develop practical guidelines on how to manage these. The guidelines were compiled in a Global IP Playbook, which can be used as an easy-to-use guide on IP management. This playbook is helpful for actors with limited experience in IP, who wish to create transparency in the different IP-related processes within the cooperation. Furthermore, it enables employees to gain knowledge useful in negotiations efficiently. To secure and guard IP interests, act as a basis for contract negotiations and division of responsibilities, and improve evaluation of IP risks are some features of the Global IP Playbook. Its primary user is Scania, and it is based on the Venture Capital (VC) investment process to enable efficient management of IP-related matters. Playbooks are common at Scania and are typically in the form of a report, where their concise structure and content allow non-expert users to assimilate key knowledge fast and simply. For example, Scania has a work playbook that outlines Scania's go-to approaches to an inclusive and attractive workplace.

According to the book *Patent management: Protecting Intellectual Property and Innovation* by O. Gassmann et al.[7], the effect of patents is strongly sector-dependent, where automotive and transport are some examples of sectors. The guidelines are therefore, in addition to the market (e.g. European Union (EU), China, and the United States (U.S.)), dependent on the industry and tech sector. Furthermore, these guidelines that make up the Global IP Playbook inform both Scania and the start-ups on how to act in the different steps of the VC investment process. They also provide an understanding of what is considered good IP protection for various technical areas and company maturity.

Chapter 2

Analytical Framework

2.1 State of Knowledge

2.1.1 Open Innovation in Start-Ups

In the book *Open innovation: Researching a new paradigm* by H. Chesbrough et al.[8], a quote by professor H. Chesbrough can be found saying: "Open innovation is the use of purposive inflows and outflows of knowledge to accelerate internal innovation, and expand the markets for external use of innovation, respectively. This paradigm assumes that firms can and should use external ideas as well as internal ideas, and internal and external paths to market, as they look to advance their technology." H. Chesbrough is known as "the father of open innovation" and has focused his research on the management of technology and innovation.[9]

Open innovation is described as having two modes: Inbound and outbound. Inbound open innovation is about external ideas or technology flowing into an organisation, meanwhile, outbound open innovation is about an organisation´s internal ideas or technology being used by another organisation that is better prepared to further develop and commercialise it. Prior studies on open innovation in the start-up scene have concluded that it is important for start-ups to engage in open innovation cooperation with incumbent firms. Start-ups engage with several external partners to gain new ideas or technologies, given their limited Research and Development (R&D) resources, in inbound innovation, meanwhile, start-ups act as technology providers, thereby an important source of innovation, to incumbent firms in outbound innovation. In Figure 2.1, the two cases of open innovation are presented in the context of a start-up and an incumbent firm.[10]

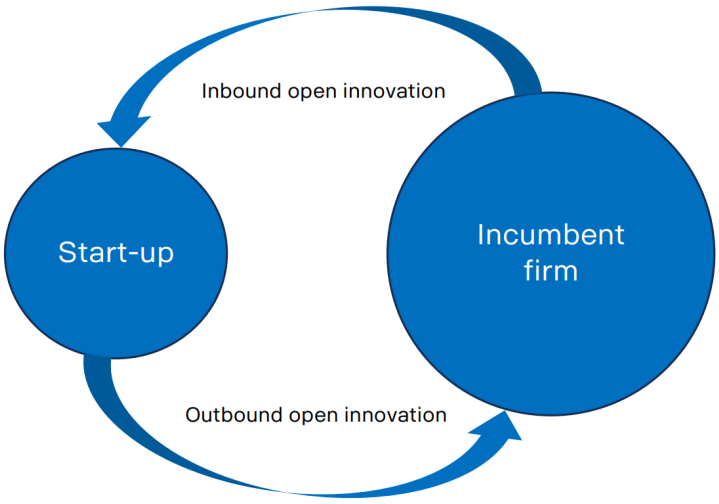


Figure 2.1: Two cases of open innovation.[10]

In the article "The Challenge of Closing Open Innovation: The Intellectual Property Disassembly Problem" by O. Granstrand et al.[11], the IP disassembly problem is addressed. This problem is about disentangling and allocating IP rights at the closing of an open innovation project or when one or more of the vital participants leave. To facilitate the management of the IP disassembly problem, it is crucial to understand what knowledge assets the partners hold and provide before, during, and after the project. These knowledge assets can be grouped into four IP types relevant for R&D collaborations: *Background Knowledge*, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*. IP relevant to a collaborative venture or open innovation project that is supplied by the partners at the start of the project is *Background Knowledge*, and all the IP produced within the collaborative venture or open innovation project during the project's tenure is *Foreground Knowledge*. Furthermore, IP relevant to a collaborative venture or open innovation project that is produced outside the project by any of the partners during the project's tenure is *Sideground Knowledge*, and IP relevant to a collaborative venture or open innovation project that is produced by any of the partners after the project ends is *Postground Knowledge*. In Figure 2.2, a representation of the relationship between the different IP types is shown (a collaboration between "Firm a" and "Firm b").[11]

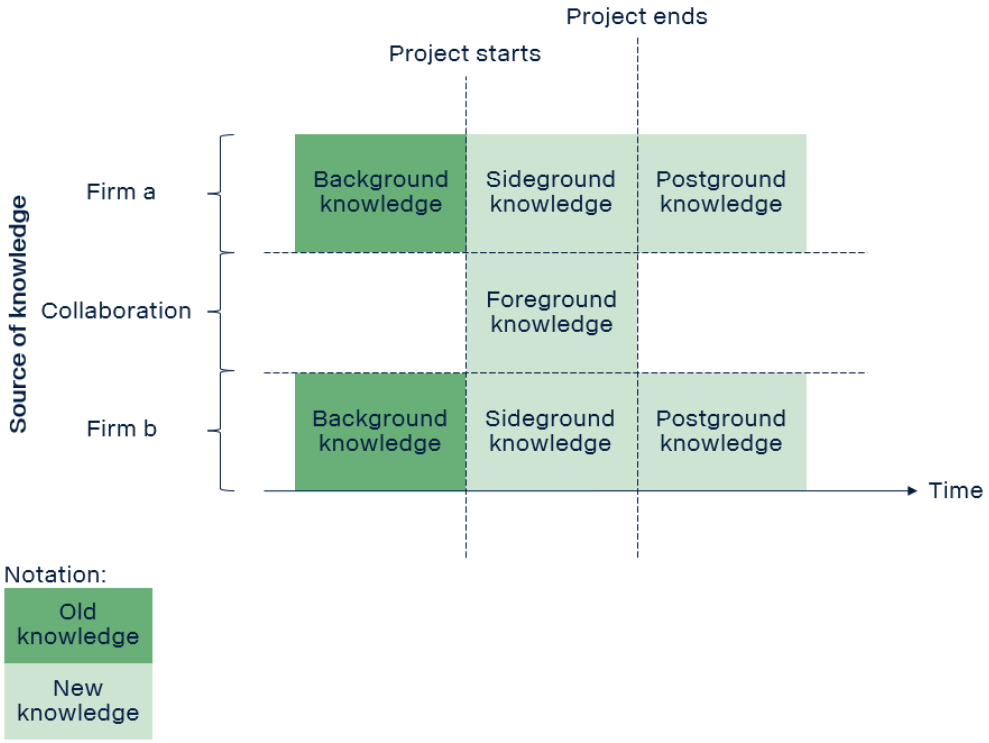


Figure 2.2: Representation of the relationship between the different IP types, inspired by O. Granstrand et al.[11]

O. Granstrand et al.[11] state that an early analysis of the case of a collaboration for joint technology development and commercialisation between a large European multinational corporation and a small U.S. R&D firm introduced the concepts of *Sideground Knowledge* and *Postground Knowledge* as complements to the standard concepts of *Background Knowledge* and *Foreground Knowledge*. Moreover, they conclude that technology and IP managers must pay more attention to *Sideground Knowledge* and *Postground Knowledge* to reduce the risks of failure in open innovation. To secure usage rights to *Sideground Knowledge* and *Postground Knowledge*, O. Granstrand et al. suggest the use of change-of-technology clauses in agreements. *Assign-Back Clause*, *Grant-Back Clause*, and *Grant-Forward Clause* are some of the most common clauses used in license agreements. *Assign-Back Clause* stipulates that ownership of improvements of the licensed technology made by the licensee must be transferred to the licensor. *Grant-Back Clause* stipulates that improvements of the licensed technology made by the licensee must be licensed to the licensor. *Grant-Forward Clause* stipulates that the licensee must be offered a license on improvements of the licensed technology made

by the licensor. Another useful clause is a *Change-Of-Control Clause*. It is used to outline how the parties are allowed to use different IP, in case a change in ownership occurs. For instance, a clause of this form could stipulate that the licensor has the right to terminate the license agreement in case of a change of ownership of the licensee.

2.1.2 Venture Capital Investment Process

In the article "Fundamentals of Venture Capital Financing and Spatial Proximity" by M. Bender [12], it is stated that corporate venture capitalists belong to the private venture capitalist category. Aiming to build up long-term strategic relationships with their investee companies and monitor new technological developments, corporations maintain CVC subsidiaries. Furthermore, corporate venture capitalists mainly make equity investments and provide frequent hands-on support. Investments are mostly done in multiple financing rounds, and portfolio companies have to achieve predefined milestones to receive further funding. This is a way of managing the high risks involved and giving the venture capitalist the option to abandon an investment if it does not develop according to the investor's expectations.

When investing CVC, there are several steps which are commonly followed. These steps can vary a little bit in shape, however, the idea is generally the same. Professors A. V. Bruno and T. T. Tyebjee have developed a model for the VC investment process, which is commonly referred to and contains six steps. These are as follows: *Deal Origination*, *Deal Screening*, *Deal Due Diligence*, *Deal Structuring*, *Investment Development*, and *Investment Exit*. In Figure 2.3, the VC investment process inspired by the model made by professors A. V. Bruno and T. T. Tyebjee is presented. The different steps are described below.[12]

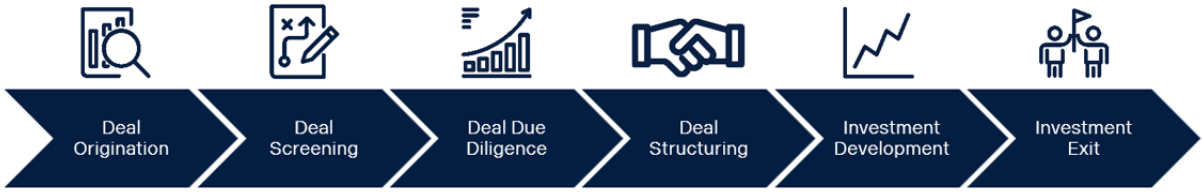


Figure 2.3: The VC investment process inspired by the model made by professors A. V. Bruno and T. T. Tyebjee.[12]

Deal Origination is the investment opportunities sourcing made by venture capital firms. To be successful, the idea is to have an adequate deal flow that is high in both quality and quantity. High quality is about having access to investment opportunities that have a high potential for value generation and offer reasonable financial terms and conditions for investing in them. These criteria need to be fulfilled for high capital gains to be possible to generate. High quantity, however, is about having the opportunity to be more selective in the deal screening and deal due diligence by having a high number of investment opportunities.

The selection of the most promising investment opportunities is one of the key tasks of venture capitalists. In the *Deal Screening*, venture capitalists apply several specific venture capitalist screens as well as a generic screen to secure certain aspects of the venture capitalist's investment strategy and efficiently reduce the number of investment opportunities. The aim of this is to receive a reduced list of deal flow that is worth further investigation. Specific venture capitalist screens include, for example, the venture capitalist's targeted investment volume, industries and technologies, the financing stage, and the geographic location of the investment. If these criteria are not met by an investment opportunity, it is eliminated from the list.

Deal Due Diligence is conducted by venture capitalists on an extensive scale after deal screening. This goes more into detail compared to deal screening, and it is performed on the remaining investment opportunities. During the deal due diligence process, venture capitalists assess the investment risks as well as the required and potential return. The goal is to gather information about, for instance, the entrepreneurial team, market potential, technology, financial needs, and legal concerns. The investment process ends at this point if the venture does not have the potential to generate the required return.

Within the *Deal Structuring*, several legal documents are negotiated. VC investments, and thus also the specific terms, are overall highly individual as ventures and their circumstances are heterogeneous. Moreover, VC investments include both financing and non-monetary support in most cases. The specific terms of the deal are reflected in a legally non-binding term sheet, and then these terms are included in a range of legal documents such as the shareholders' agreement, the articles of association, the by-laws of the management and supervisory board, and the employment contracts of the management. The bargaining power of the entrepreneurial team and the

venture capitalists affects the outcome of the negotiations. Deal structuring is a critical step, firstly because an appropriate deal structure is crucial for the venture capitalists to earn the target rate of return. Secondly, the foundation of future cooperation between the venture capitalist and the entrepreneurial team is represented by the negotiated terms in the deal structuring.

The *Investment Development* starts after the VC financing contract is signed, in which venture capitalists monitor and support their portfolio companies. By conducting these activities, venture capitalists aim to protect and increase the value of their portfolio companies. More specifically, the aim is to minimise risks and maximise returns of the VC investments. Support from the venture capitalists could be in the form of financial help, to serve as a sounding board to the management, participate in strategy development, provide feedback to the management, help the management in operational aspects, and recruit management.

A certain time after the venture capitalists´ initial investment, the venture capitalists intend to sell the shares of their portfolio companies. The *Investment Exit* is vital for their return generation, as they generate most of their investment returns in the form of capital gains. Moreover, the venture capitalist and the entrepreneurial team have to determine the point of time for an exit as well as the exit channel. These decisions offer a high potential for conflicts since both parties might have different interests. Therefore, it is important to discuss potential exit strategies as early as possible and consider the interests of all involved parties. When preparing the investment exit, activities involved are, for instance, identification of potential purchasers, improvement of the venture's accounting and reporting systems, evaluation of the venture to determine a range for the sale price, and assembly of information to enable potential buyers to evaluate the venture.

2.1.3 General Intellectual Property Challenges

There are general IP challenges to be dealt with that, for instance, apply in the corporate venturing context. These must be treated carefully for the cooperation between the incumbent firm and start-up to work properly. The general IP challenges to be considered are the following: *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*. These challenges are connected to the IP types explained by O. Granstrand et al.[11]. With inspiration from the representation in Figure 2.2 and an interview with M.

Holgersson, co-author of the article from which the representation originates, Figure 2.4 is produced. Compared to Figure 2.2, this figure investigates an "Incumbent firm" and a "Start-up" instead of "Firm a" and "Firm b". Figure 2.4 applies in the VC investment process (see Figure 2.3) by using the terms *Deal Structuring*, *Investment Development* and *Investment Exit* to describe when each IP type is relevant throughout the cooperation. Both the incumbent firm and start-up provide *Background Knowledge*, which is determined during the *Deal Structuring* and used from the start of the *Investment Development*. The *Foreground Knowledge* is defined as the IP arising from the start-up´s R&D during the *Investment Development*, and there is no *Sideground Knowledge* for the start-up. The reason for this is that the collaborative research activities are performed in the start-up´s business. The CVC firm representing the incumbent firm invests capital, resources, and strategic management in the start-up and its development work, meaning the start-up is highly dependent on the incumbent firm. While maintaining resources for the start-up, the incumbent firm can develop *Sideground Knowledge* useful for the cooperation. After the *Investment Exit* is executed, it is possible that *Postground Knowledge* arise from both the incumbent firm and start-up, which is relevant for the cooperation.

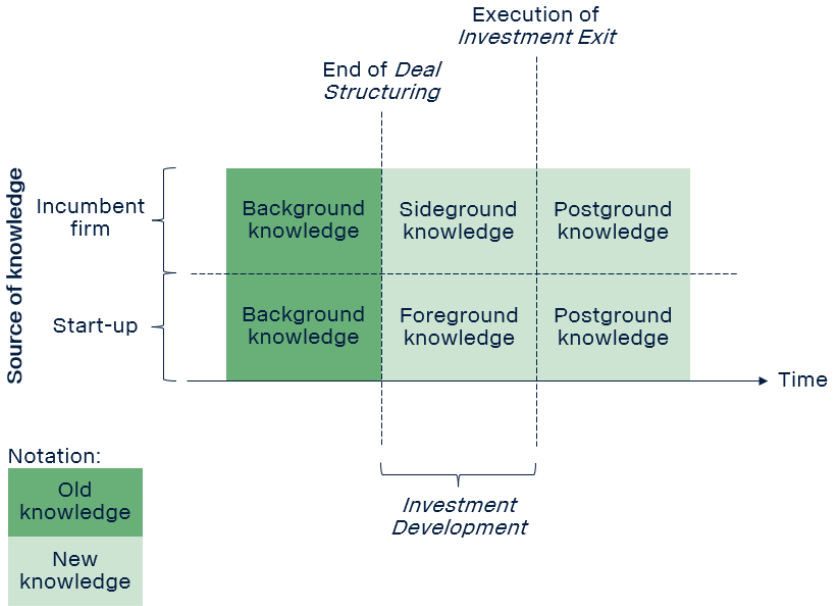


Figure 2.4: Representation of the relationship between IP types that applies in the VC investment process, inspired by O. Granstrand et al.[11] and an interview with M. Holgersson.

Below, a walkthrough of the general IP challenges *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage* is given. In this walkthrough, these are explained and connected to the IP types and the VC investment process.

Fenwick is a premier law firm for tech and life sciences companies that, for instance, guide clients through all stages of their lifecycle to create, monetise, manage, and protect their IP assets. Moreover, they are operating in the landscape of corporate venturing, and their clients are both start-ups and market leaders. This law firm states that CVC firms should evaluate the *IP Strength* of a start-up´s patent portfolio and its know-how (i.e. the knowledge and skills needed to implement inventions) to determine if it is worth investing in a start-up or if it is easy for others to simply replicate their technology.[13] To evaluate the start-up´s routines is necessary. Before investing in a certain technical field, it is crucial to perform an IP landscape analysis. This is a comprehensive analysis and evaluation of the IP terrain in a certain tech sector, industry, or market that is performed early in the development process. For the identification and assessment of the existing IP rights and their potential impact on business or research activities, this analysis involves a systematic search and review of patents, patent applications, scientific literature, market reports, and other relevant sources.[14] According to Fenwick, CVC firms should also check if a Freedom To Operate (FTO) analysis is conducted by the start-up, i.e. an investigation of the possibilities for a product or service to be introduced on the market without the risk of being hampered by others´ rights.[13] FTO is about being able to use a product or process without infringing on existing IP rights.[15] Evaluation of a start-up´s routines (e.g. IP landscape and FTO), know-how, and patent portfolio should be performed during the *Deal Screening* and *Deal Due Diligence* on *Background Knowledge*.

Fenwick explains how there are several challenges involved in securing good conditions for *IP Ownership* in corporate venturing. For instance, CVC firms´ interest in ensuring that the start-up has a clear title to its IP. This means there are no third-party ownership claims.[13] If there are several investors in a start-up already, the ownership of the start-up´s IP may be shared between involved parties (i.e. the start-up and its investors). A further risk is that the right to inventions has not been correctly assigned from all inventors to the start-up, resulting in unclear ownership of the inventions and thereby also any ensuing patent applications. This needs to be considered by CVC firms when

determining whether or not to invest in a start-up. During the *Deal Screening* and *Deal Due Diligence*, the CVC firms do some research on the start-up to see if it has a clear title to its IP. Here, the targeted IP is *Background Knowledge*. The terms for the use of this IP are negotiated in the *Deal Structuring*.

World Intellectual Property Organisation (WIPO) leads the development of a balanced and effective global IP ecosystem to promote innovation and creativity for a better and more sustainable future. This organisation explains the importance of *IP Rights Assignment*. One case is the way of regulating the rights assignment of new IP arising within the cooperation between the incumbent firm and start-up. This covers *Sideground Knowledge* and *Foreground Knowledge* that is developed during the *Investment Development*.[16] One approach possible for managing these is the use of something similar to a Joint Development Agreement (JDA). A JDA determines the terms for two or more parties working together to develop a particular technology, and is typically negotiated before or during the *Investment Development*. IP is central in cooperations covered by this type of agreement, therefore, the JDA should clearly state who owns what.[17] It is appropriate to manage this during the *Deal Structuring*, since IP that arises during the *Investment Development*, i.e. *Foreground Knowledge* and *Sideground Knowledge*, is highly critical. There also needs to be guidelines for rights assignment of *Postground Knowledge* that arises after the *Investment Exit*. The IP rights assignment is of main importance, since it determines which party has control of, for instance, licensing. One potential risk is conflict of interest, for instance, that the start-up needs revenue streams and therefore wants to license IP or sell it (i.e. transferring all exclusive rights to a patented invention) to parties outside the cooperation, meanwhile, the incumbent firm wants to avoid this to further develop the IP on their own. To manage IP rights assignment at an early stage in a way that satisfies everyone is necessary to avoid conflicts during the cooperation.

FasterCapital is a global venture builder and online incubator dedicated to co-funding and co-founding innovative startups. This venture builder states that the strategic management of *IP Licensing* is essential for a start-up when defining the trajectory of its growth and success in the competitive landscape of start-ups. The broadness and depth of a start-up's license scope can either lead to expansive opportunities or restrict it to a narrow market segment. To ensure that the license scope aligns with the start-up´s long-term business objectives and market potential, it is important to evaluate and

optimise the license scope. Furthermore, for a start-up to achieve scalability and flexibility in its licensing, it is beneficial for them to promote scalable license agreements. This would allow for expansion without the need to renegotiate terms.[18] When negotiating with CVC firms, start-ups must keep the scope of the licensing agreements in mind. The negotiations between the parties must be in such a way that everyone agrees on the terms. The license agreements are preferably negotiated during the *Deal Structuring*. Here, *Background Knowledge* is managed, and the licensing framework for *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge* is developed.

Fenwick explains how CVC firms have an interest in investigating what measures the start-up has taken to protect core IP assets and avoid *IP Leakage*. This could, for instance, refer to the management of the patent application of innovative technology and the safeguarding of trade secret information. Furthermore, CVC firms often want to ensure that a start-up appropriately stores, monitors, and restricts access to source code as well as other valuable data. Sensitive business and technical information should be protected by a Non-Disclosure Agreement (NDA) and disseminated responsibly.[13] NDAs are legally enforceable contracts that form the basis of a confidential relationship between someone who possesses sensitive information and someone who will gain access to that information. A confidential relationship means both parties have a duty not to share that information in violation of the terms of the NDA.[19] To ensure each party in the cooperation takes responsibility for information protection, NDAs should be managed already in the *Deal Structuring*. Information to be protected involves *Background Knowledge*, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*.

2.2 Research Questions

The literature review highlighted a gap in the research on IP challenges in corporate-backed ventures. Firstly, this was identified when the representation of the relationship between IP types made by O. Granstrand et al.[11] (see Figure 2.2) was studied. It applies to collaborations in general, and not uniquely to cooperations in the corporate venturing context. Secondly, a gap in the research was noted when the general IP challenges in relation to the VC investment process were explored. Here, the main focus was on IP in the *Deal Screening* and *Deal Due Diligence*, meaning that IP in the *Deal Structuring*,

Investment Development, and after *Investment Exit* were a bit left out. The general IP challenges did not cover all IP challenges involved in corporate venturing, meaning an investigation of IP challenges in the different steps of the VC investment process was needed. Therefore, the purpose of this master thesis project was to identify key IP challenges in corporate-backed ventures and develop practical guidelines on how to manage them. To achieve this purpose, the following research questions were investigated:

1. What key challenges are encountered when managing IP in corporate-backed ventures?

2. What practical guidelines are needed for fruitful management of key IP challenges in corporate-backed ventures?

Chapter 3

Research Design

3.1 Delimitations

This study was directed to CVC. Although the context of the study was similar to the Mergers and Acquisitions (M&A) situation (mergers refer to two companies combining to create a new single entity, meanwhile, acquisition refers to a company acquiring another one.[20]), the time constraints of the project prompted me to only focus on CVC. However, many of the findings could be relevant to the M&A situation as well. Regarding the Global IP Playbook, a lot of work was put into the core content. There could be other information that can enhance the relevance of the playbook for firms, but that is left to future scholars.

3.2 Research Setting

Scania is a world-leading provider of transport solutions. It was founded in 1891 and is a manufacturer of commercial vehicles and power solutions (e.g. diesel engines).[21] Explained on a basic level, a vehicle is commercial when a business registers the vehicle as such and uses it for business purposes. This can, for instance, be a car or a truck.[22] Scania had 59 397 employees in Q3 2024, and the net sales in MSEK of that quarter were 48 228 (to be compared with 47 372 MSEK in Q3 2023). Their purpose is to drive the shift towards a sustainable transport system, creating a world of mobility that is better for business, society, and environment.[21] In its journey towards sustainable transport, Scania is part of strategic collaborations. Scania finds strategic collaboration opportunities, for instance, by being part of the Combient network (Combient Foundry) to connect with start-ups. This is a Nordic-based

industry cooperation aiming at accelerating transformation in tech sectors such as automation, Artificial Intelligence (AI), analytics, and open innovation. The Combient Foundry is an initiative that enables the joint creation of new services and solutions by linking start-ups with Nordic industry leaders.[23]

This master thesis project, which was a case study conducted in collaboration with Scania CV AB, was hosted by Scania´s IP department. In addition to people at Scania IP, interactions were made with people at Scania Ventures and New Business (VNB) and Scania R&D. Scania IP, R&D, and VNB are all parts of the innovation network at Scania. All these departments contribute to the development of world-class transport solutions. While Scania R&D focuses on in-house development, Scania VNB has its gaze into the future, scouting for the groundbreaking solutions of tomorrow. To know which start-ups to initiate cooperation with, i.e. to have an idea of which start-ups provide value to Scania R&D, evaluation and assessment of the start-ups' business ideas and products are needed. This is made by Scania VNB, where they investigate and compare start-ups that could be of interest. There is a dialogue between Scania R&D and Scania VNB about the different cases, where the business and technological aspects are frequently discussed to achieve clarity.

To remain at the cutting edge of technology and drive development within the industry, Scania puts a lot of effort into its R&D department. Sustainable solutions and reduced climate impacts are the main focus, and to meet demands from the customers and community, every detail within design and construction is continually being improved. Innovation is encouraged within R&D, with the vision to provide solutions for the future. New solutions are developed with regard to customers´ needs, the evolving environment, and competitive context. All the R&D efforts are a strong commitment to quality. The whole chain, from the underlying research, construction and testing through to quality assurance within production and in customer vehicles in service on the world's roads, is covered by the area of Scania R&D. The aim is to maintain a broad competence to ensure that Scania retains its leading position within the commercial automotive industry.[24] To protect the outcome of large investments made in cutting-edge technology, Scania´s IP department has throughout the years accumulated competence and developed processes for the protection of in-house developed IP. This is, for instance, patents, designs, and trademark-related matters.

The mission at Scania VNB is "accelerating the shift towards a sustainable transport system by developing complementary and transformational businesses that strategically position Scania at the forefront of the evolving transport and logistics ecosystem".[25] Their strategic direction for building new growth is defined by the following four areas: *Energy & Infrastructure*, *Asset Management & Circularity*, *Autonomous & Supply Chain*, and *Future-Tech Enablers*. *Energy & Infrastructure* supports the evolution of the transportation industry, and *Asset Management & Circularity* combines new ideas in materials, recycling, and eco-friendly designs to make the industry greener. *Autonomous & Supply Chain* plays an increasingly important role in steering the industry towards a prosperous, sustainable, and digitally enhanced transport and logistics ecosystem, and *Future-Tech Enablers* involves areas such as cybersecurity, semiconductors, quantum computing, and robotics that are becoming essential in the shift towards a sustainable future. Scania Invest is the venture capital unit within Scania VNB. In addition to providing capital to their portfolio companies, Scania Invest drives value and serves as a long-term partner to these. Moreover, they strive to create a sustainable impact in the transportation ecosystem by supporting innovative technologies and visionary entrepreneurs. They believe in the strength of partnerships, where diverse expertise and experiences are combined to drive stronger outcomes. Scania Invest supports its portfolio companies by leveraging Scania's internal capabilities with its hands-on operational legacy.[26]

To facilitate strategising investments, Scania VNB has defined three "Horizons". These are presented in Figure 3.1, and are as follows: *Horizon 1: Core Business Optimisation & Incremental Innovation*, *Horizon 2: Emerging Growth & Business Expansion*, and *Horizon 3: Future Disruptive Innovation & Market Creation*. In this figure, there is *Time* on the x-axis and *Value* on the y-axis. The horizons are a way of managing investment opportunities based on their level of innovation. *Horizon 1: Core Business Optimisation & Incremental Innovation* has "low" value and is close in time. This horizon covers projects important to Scania´s core business. The projects could, for instance, originate from Scania R&D. *Horizon 2: Emerging Growth & Business Expansion* has "medium" value and is further away in time. For instance, projects within this horizon are highly related to electrification. *Horizon 3: Future Disruptive Innovation & Market Creation* has "high" value and is far away in time. This horizon covers highly complex projects, such as quantum computing and AI.

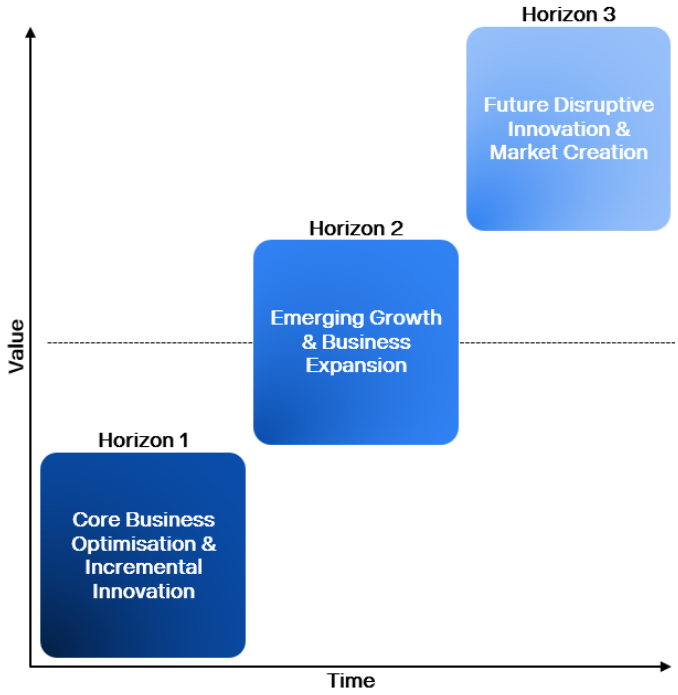


Figure 3.1: The three horizons of innovation defined by Scania VNB.

Incremental innovation, which is covered by *Horizon 1: Core Business Optimisation & Incremental Innovation* in Figure 3.1, is a series of small improvements or upgrades made to a company´s existing products, services, processes or methods. Meanwhile, disruptive innovation, which is covered by *Horizon 3: Future Disruptive Innovation & Market Creation* in Figure 3.1, is when a company with fewer resources moves upmarket and challenges an incumbent firm´s business. There are two types of disruptive innovation:

- *Low-End Disruption*: To enter at the bottom of an existing market and claim a segment, a company uses a low-cost business model.
- *New-Market Disruption*: To create and claim a new segment in an existing market, a company cater to an underserved customer base.

The IP challenges that Scania VNB encounters through its interactions with start-ups differ significantly from those arising from Scania R&D´s in-house operations. Since Scania IP's activities are primarily tailored to support Scania R&D, there is limited experience within the organisation with the specific IP issues that arise in Scania VNB's context. The research questions are designed to address this gap, and the findings of this study are presented in a Global IP Playbook to better integrate IP considerations into Scania VNB's processes.

3.3 Research Method

The first step of the master thesis project was to choose a topic. Here, there were several candidates. After discussions with the supervisors and manager about the different candidates, everyone agreed that *Intellectual Property Challenges in Corporate-Backed Ventures* was the most interesting one to proceed with. The inspiration for this project came from Scania VNB, which had felt for some time that easily accessible information about IP challenges in their operations was lacking. In this project, a Global IP Playbook was developed. Here, people's IP-related experiences, ideas, and reflections were interesting to investigate. This is because these experiences highlighted different IP challenges, which made it possible to develop practical guidelines for fruitful management of them. In this case study, an abductive approach (abductive reasoning) was used. There are many benefits of conducting research in this way, therefore, this approach was applied in this master's thesis project. In-depth semi-structured interviews were conducted, meaning that a series of participants were interviewed one-on-one to explore patterns. Here, a research discussion guide was prepared beforehand to guide the conversation and, at the same time, allow flexibility in the interview. The research discussion guide is presented in Figure A.1.

Case studies focus on either one instance or a few instances of a certain phenomenon. This is done with a view that provides an in-depth account of experiences, relationships, processes, or events that occur in that particular instance or those particular instances. For case studies, the aim is to illuminate the general by looking at the particular. Having the possibility to use a variety of methods, depending on the circumstances and specific needs in a certain situation, is one of the case study's strengths.[27] To have an abductive approach in the case study means that the reasoning is a creative process suitable for complex problem-solving. The processes and output are always transparent and explainable.[28] According to the paper "Validation in Interpretive Management Accounting Research" by K. Lukka et al.[29], what characterises abduction is that it relies on the development of theoretical reasoning by combining empirical and theoretical insights about the phenomenon under observation.

The abductive reasoning starts with *Observation*, continues with making an *Assumption* and creating a *Hypothesis*, and ends with *Best Guess* through validation (i.e. accepting or rejecting) of the hypotheses by conducting experiments. In Figure 3.2, an overview of the abductive reasoning process is presented.[30]

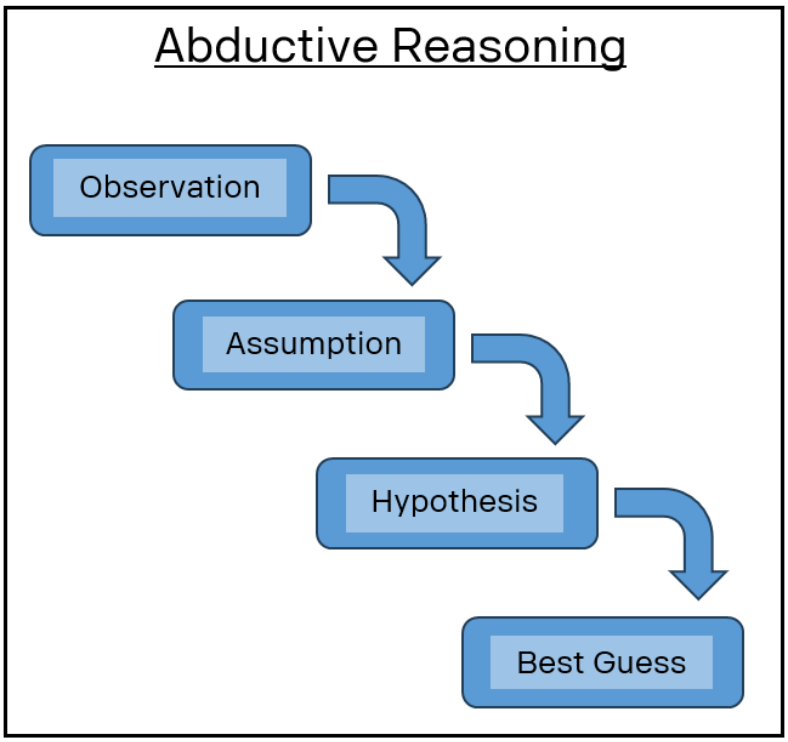


Figure 3.2: Overview of the abductive reasoning process, inspired by [30].

The investigation started with *Observation*. It was observed that Scania VNB felt a lack of easily accessible information about IP challenges in their operations. To provide Scania VNB with more information about IP in its processes, an investigation of IP challenges in corporate-backed ventures was conducted. A literature review was performed to see what the research covered, and semi-structured interviews were conducted to explore the lived experiences of respondents. Throughout the project, there was jumping back and forth between the literature and interviews. This means that knowledge gained from the literature was brought to the interviews, and then new insights from the interviews were brought back to the literature. This

procedure continued in this way, meaning that the literature and interview data collection strengthened each other. The literature review was performed to gain knowledge about the research topic and identify known IP challenges. Meanwhile, the semi-structured interviews were conducted to validate the literature findings and find new findings which contribute to the research. To keep the interview discussions inside the scope of the master thesis project, the research discussion guide found in Figure A.1 was used as a basis. In *Assumption*, initial explanations were developed with the use of the literature review and semi-structured interviews. These were then refined to testable propositions in *Hypothesis* and further explored. The three steps mentioned so far, *Observation*, *Assumption*, and *Hypothesis*, resulted in concepts. These were IP structures as well as key quotes collected from the semi-structured interviews. The concepts were used in *Best Guess*, in which key IP challenges were identified. They acted as a basis for the development of practical guidelines used for fruitful IP management.

3.4 Data Collection

The respondents were selected to represent a broad set of perspectives. This was to gather data which captures the interests and opinions of all parties involved as much as possible. These had varying backgrounds, meaning that they, for instance, differed in work experience. Interviews and meetings were not only conducted internally, but also on an external level. People from companies other than Scania contributed by sharing their knowledge and experience within the research field. Respondents represented roles such as postdoctoral researcher (university), director of patent unit (IP), development engineer (R&D), and investment associate (VC and M&A). Through the supervisor at Scania, Mimmi Hanson - Senior Patent Expert, patent professional, and researcher with a Doctor of Philosophy (PhD) from the Stockholm School of Economics, and the manager at Scania, Douglas Elliot - Head of IP Enforcement and Strategies responsible of patent licensing and competitor surveillance matters, the names of many of the respondents were gathered. Then, some names were given by respondents. Details about the respondents interviewed are presented in Table 3.1

Table 3.1: Details about respondents interviewed.

Respondent	**Role/Responsibility**	**Company/University**
A	Postdoctoral Researcher, Department of Technology Management and Economics	University X
B	Director, IP Support	Company 1
C	Director, Patent Unit	Company 2
D	Group Manager, Data Streaming	Scania
E	Development Engineer	Scania
F	Head of Litigation & IP	Scania
G	Partnership Manager	Scania
H	Senior IP & Technology Advisor	Company 3
I	Partnership Manager	Scania
J	Innovation Office	Scania
K	Investment Associate, VC & M&A	Scania
L	Investment Associate, VC & M&A	Scania
M	Vice President, Head of IP	Company 4
N	Patent Manager	Company 5
O	Head of Legal	Company 6

The semi-structured interviews were conducted with the use of Microsoft Teams. The benefit of this approach was that it was possible to use Teams for transcribing. This made the data collection easy and efficient. In near real time, live transcriptions showed speech-to-text of spoken content during Teams meetings. The text, with the speaker's name and a time stamp, appeared alongside the meeting video. After the meetings, the transcriptions were

downloaded. These transcriptions were compared with the audio recording to ensure that the transcriptions were correct. The transcriptions were translated from Swedish to English. Another benefit of Teams was that it enabled meetings over any distance. This option was very convenient because it allowed participants from locations outside the Stockholm area.

3.5 Data Analysis

AI tools were used to support the work and allow efficient data analysis, which was good because there were time constraints. AI was used as a tool to analyse the transcripts created during the semi-structured interviews. The benefit of using AI was that it pointed out key aspects from each interview, which made it easier to know what to look for when studying and coding the data. AI was also applied in the correction of spelling and grammar within this master thesis report, where the AI writing assistant *Grammarly* was used.

The data was analysed following the Gioia method, highlighting critical IP challenges to consider in cooperation between an incumbent firm and a start-up. The data structure started with concepts based on the coding of the data, continued with themes developed from these concepts, and ended with *Aggregate Dimensions* representing key IP challenges.

3.6 Validity Assessment and Generalisability

The data collected throughout the study were validated in different ways. The research started with exploring the existing knowledge within the research area, where literature (i.e. books and scientific articles) was studied. Knowledge considered relevant for this project was compiled in **Chapter 2. Analytical Framework**. This acted as a basis for the case study conducted. As shown in Table 3.1, experts in IP, VC, M&A, and R&D were interviewed to ensure that data were collected from respondents with broad experience within the research area. Every other week, there were group meetings with the KTH supervisor and other students taking the master thesis course. Here, thoughts and ideas were shared between everyone to help each other progress. In addition to these meetings, there were regular meetings with the industrial supervisor. In these meetings, the progress made was followed up to ensure that the project was moving in the right direction.

This master thesis project was conducted at Scania, a heavy commercial vehicle manufacturer. Even though the project was conducted in this context, some of the findings could apply to other types of branches or companies where IP is of importance. The semi-structured interviews were conducted with external parties as well, meaning that input to the project also came from parties operating in different sectors. The key IP challenges were found in the VC context for cooperation between incumbent firm and start-up, where practical guidelines for fruitful IP management were developed and compiled in a Global IP Playbook. There is generalisability in the sense that the challenges identified and the management of these apply to this type of cooperation, which is possible to have in a wide spectrum of tech sectors, industries, and markets.

Chapter 4

Derivation of Key Intellectual Property Challenges

4.1 Key Intellectual Property Challenges Derived From Intellectual Property Structures

With the results from the research made by O. Granstrand et al.[11] and the interview with M. Holgersson as a basis, the idea behind representations of the relationship between IP types in corporate venturing was further developed. Through input from respondent K, i.e. an investment associate VC and M&A at Scania, several possible representations of the relationship between IP types in corporate venturing were created. Here, these representations are referred to as "IP structures". The input from respondent K strengthened the concept of IP structures because the input came from someone involved in the cooperation between the incumbent firm and start-up.

As an investment associate, respondent K had experience with all steps of the VC investment process: From *Deal Origination* to *Investment Exit*. Strong presence in steps such as *Deal Structuring* and *Investment Development* meant respondent K had high awareness of the negotiation process and outcomes with start-ups, both before and after initiating cooperation. A main contribution to the concept development for IP structures from the discussions with respondent K was that a change in cooperation form during the *Investment Development* could occur, entailing that the *Investment Development* could contain multiple IP structures and not only one as presented in prior literature.

There are several ways of conducting cooperation between an incumbent firm and a start-up in corporate venturing. This means that in addition to Figure 2.4, other IP structures can represent the cooperation form. For example, it is possible to have the one presented in Figure 4.1 where the cooperation has the form of a project. Here, there is both *Sideground Knowledge* from outside the project and *Foreground Knowledge* within the project throughout the *Investment Development*. In a project of this kind, the incumbent firm and start-up combine their knowledge and technologies to develop something new. This form of cooperation could, for instance, be regulated with a JDA. One benefit of choosing this cooperation form is that it enables both parties to cooperate but still conduct R&D on their own (*Sideground Knowledge* for both).

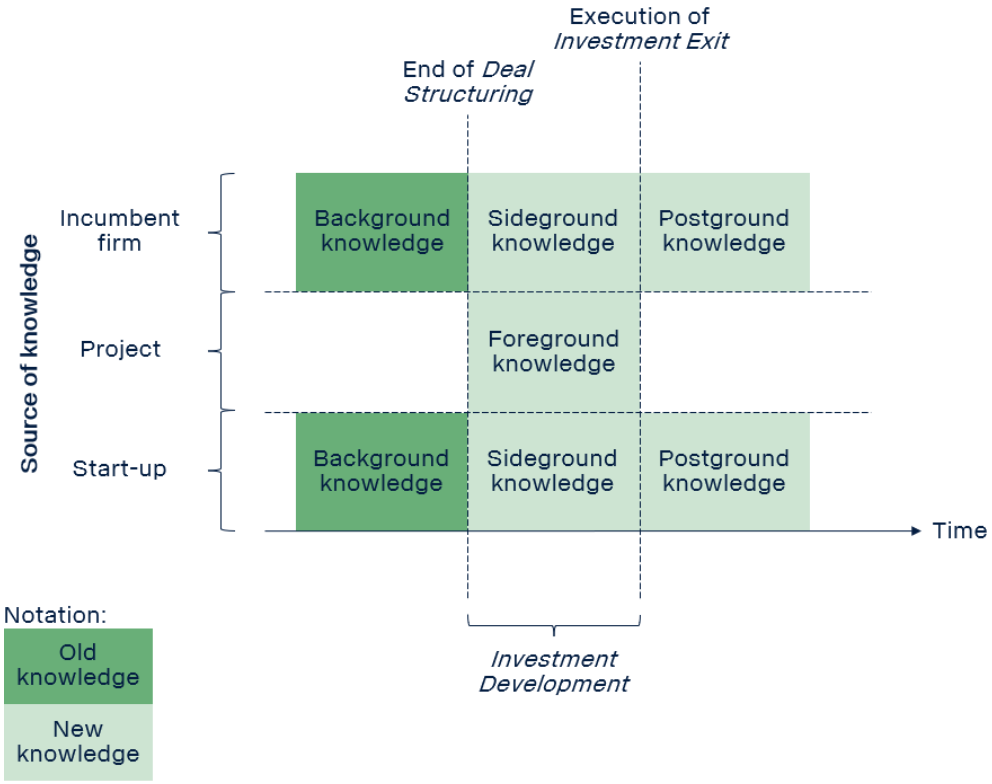


Figure 4.1: IP structure in corporate venturing suitable for cooperation with the form of a project.

Another example is presented in Figure 4.2, which represents a cooperation without any *Foreground Knowledge*. Throughout the *Investment Development*, there is only *Sideground Knowledge* that each party contributes to the cooperation. The expected outcome then consists of dialogues between the parties, where new perspectives, insights, and ideas useful for further development are shared.

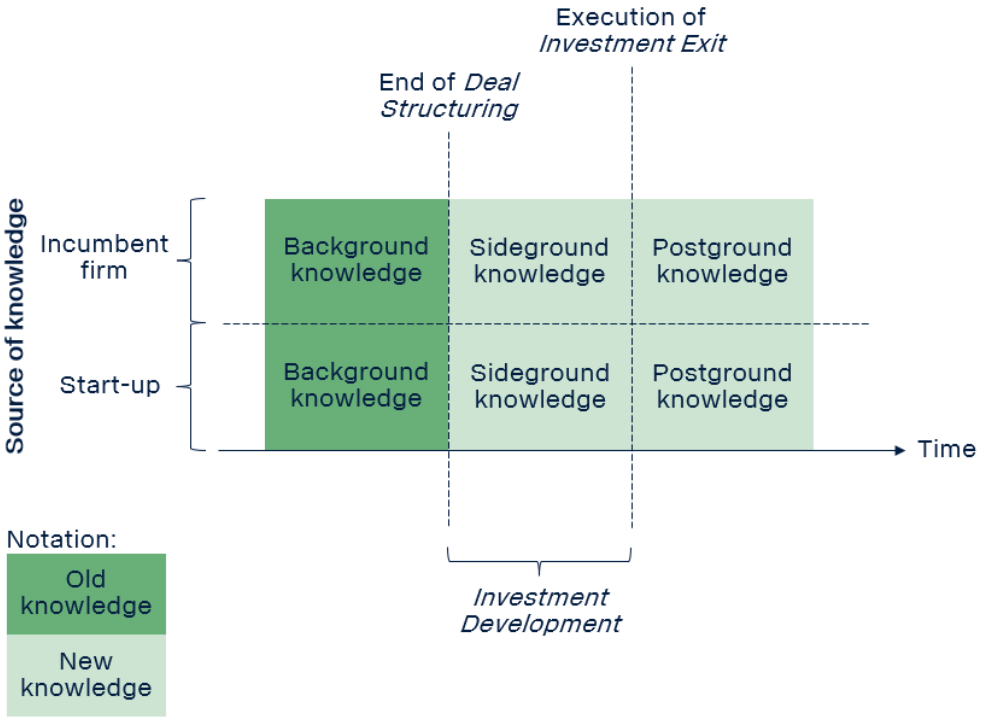


Figure 4.2: IP structure in corporate venturing that offers simplicity in the cooperation.

It is also possible to have a combination of cooperation forms. One is presented in Figure 4.3. The *Investment Development* starts with only *Sideground Knowledge* (see Figure 4.2), and then there is a change in the cooperation form where the involved parties enter a project (see Figure 4.1). This change in cooperation form means there is both *Sideground Knowledge* and *Foreground Knowledge*. The benefit of this cooperation form is that it allows both parties throughout the entire *Investment Development* to contribute with *Sideground Knowledge* (i.e. having the opportunity to perform R&D on their own).

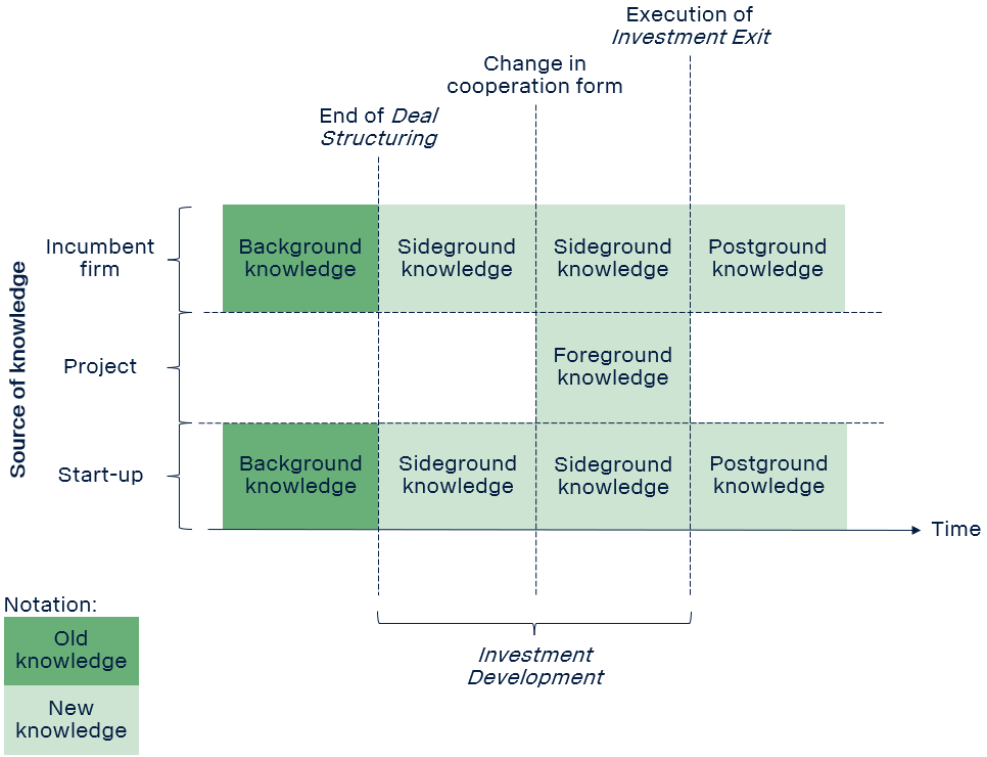


Figure 4.3: Combination of IP structures, allowing *Sideground Knowledge* for both parties.

Another combination of cooperation forms is presented in Figure 4.4. Here, the *Investment Development* is initiated with *Sideground Knowledge* from the incumbent firm and *Foreground Knowledge* within the R&D of the start-up (see Figure 2.4). After this, there is a change in the cooperation form. This change means that the cooperation goes in a different direction, instead of *Foreground Knowledge* at the start-up´s R&D there are such activities in a project (see Figure 4.1). Furthermore, a consequence of this transformation is that the start-up starts to contribute with *Sideground Knowledge* to the cooperation. A reason for making this change could be that the start-up wants the opportunity to perform R&D on their own (*Sideground Knowledge*).

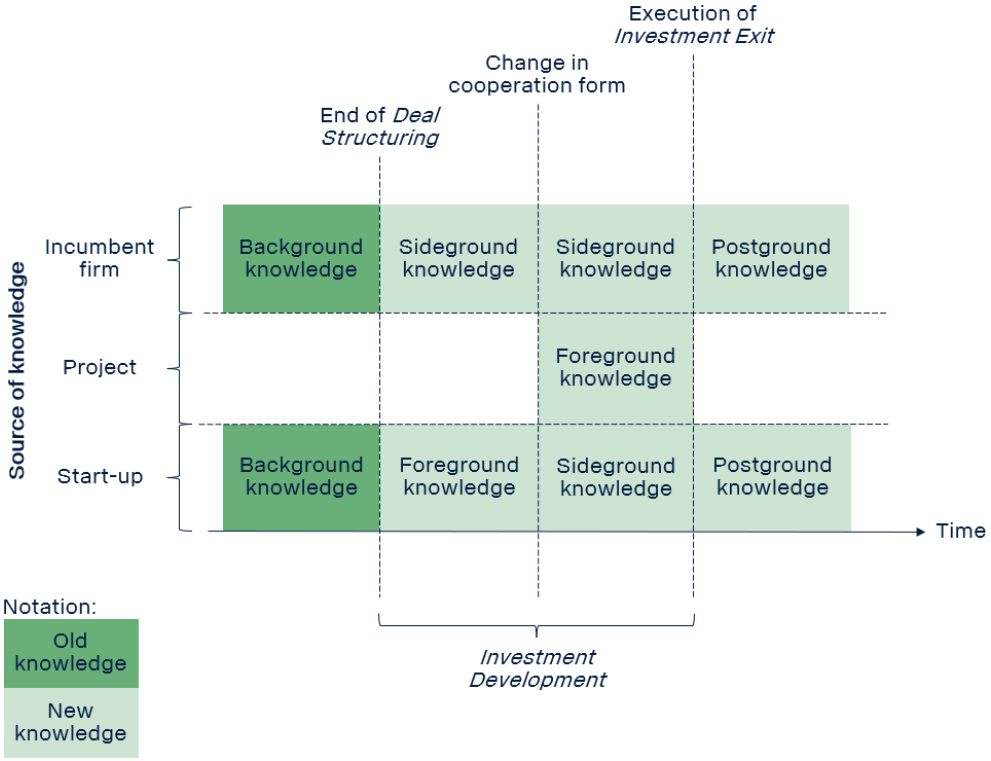


Figure 4.4: Combination of IP structures, where there is a focus on *Foreground Knowledge*.

Figure 4.5 shows a third combination of cooperation forms. In this combination, there is only *Sideground Knowledge* at the beginning of the *Investment Development* (see Figure 4.2). After this, there is a change in the cooperation form where there is *Foreground Knowledge* arising from the start-up´s R&D instead of *Sideground Knowledge* (see Figure 2.4). This type of cooperation can be of interest for the incumbent firm if they want to initiate cooperation with the start-up carefully. This means that the cooperation starts lightly with dialogues between each other, where new ideas are being shared, and later continues with more advanced cooperation. Here, the incumbent firm supports the start-up´s R&D with resources and strategic management for the development of *Foreground Knowledge*.

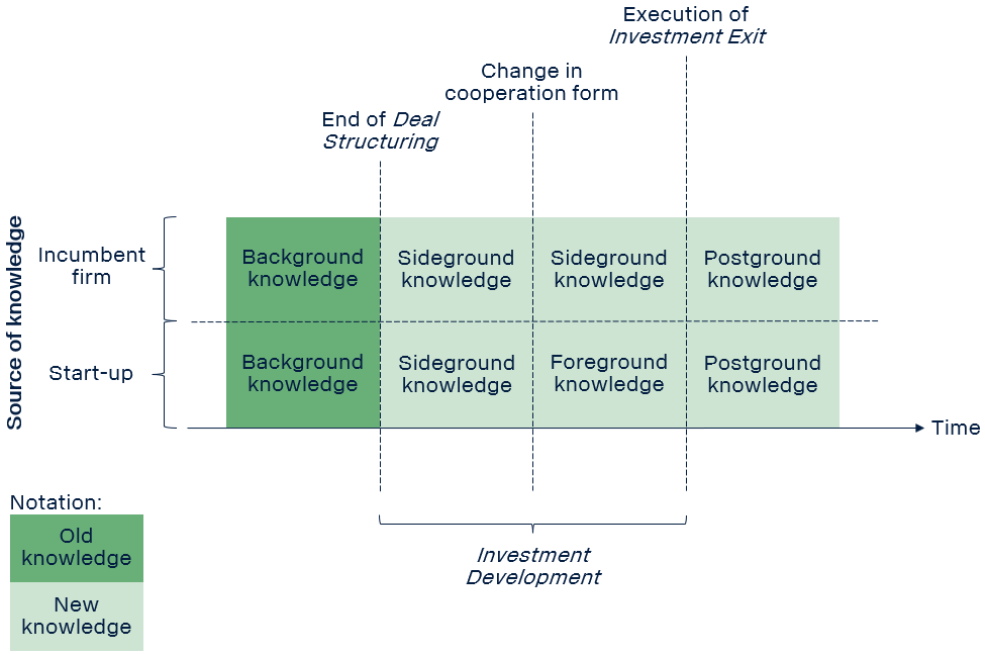


Figure 4.5: Combination of IP structures. Here, cooperation in the form of a project is avoided.

Important to note is that in addition to the combinations of cooperation forms presented in Figures 4.3, 4.4, and 4.5, there are other alternative combinations. For instance, it is possible to switch places between the two IP structures inside the *Investment Development*. Another way of changing the cooperation form is by adding new parts to the *Investment Development*. This can be done either by adding an IP structure that has already been used or by adding a new one. The reason for doing any of these implementations could be that the parties of the cooperation form want to go back to an IP structure that has been used before, because they realise that one is better. It can also be that they want to try a new cooperation form, to see if that one is more aligned with the visions and goals of the cooperation.

Based on these IP structures, several key IP challenges were identified. Moreover, they need to be considered and managed carefully for efficient cooperation between the incumbent firm and start-up. They are as follows: *Complexity in patent strategy development*, *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, and *Requirements on negotiated agreement*.

4.1.1 Complexity in Patent Strategy Development

Depending on whether the IP produced within the *Investment Development* is *Sideground Knowledge* or *Foreground Knowledge*, if the *Foreground Knowledge* arises from a project or within the start-up´s R&D, etc., the complexity of the patent strategy varies. For all parties involved to be on the same wavelength in the development of a patent strategy can be challenging. There could be different opinions about patent coverability, i.e. which markets to obtain IP protection in and what technology to protect. Another patent strategy challenge that could arise is to agree on which way to protect the IP. One party may advocate patenting, meanwhile, the other party suggests keeping the invention a trade secret.

When choosing a patent strategy, there is a risk that a conflict of interest arises between the incumbent firm and start-up. It could be that the incumbent firm expects it to be easy for competitors to circumvent patent protection or that it is hard to prove patent infringement, and therefore suggests that the *Foreground Knowledge* should be kept a trade secret. This might not align with the start-up's interest, which instead wants to secure a patent on the *Foreground Knowledge* before licensing it to other parties outside the cooperation to obtain revenue streams. The start-up could be highly dependent on revenue streams from the *Foreground Knowledge* due to its small size and unestablished business, meanwhile, the established incumbent firm could prefer to avoid licensing to competitors for its benefit.

4.1.2 Deviation From Set Plan for the Cooperation

One challenge, which could occur during the *Investment Development*, is that changes need to be made to the set plan for the cooperation. There is a risk that the structure that the parties have chosen for their cooperation no longer meets their needs. Here, it is beneficial to highlight in an early stage of the *Deal Structuring* that several changes in the cooperation form are possible. Moreover, the management involved in such changes becomes less complicated if both the incumbent firm and start-up are prepared for it. One preparation which can be done is to jointly develop contingency plans to prepare for changes in the cooperation form. For instance, one part of the plan could contain suggestions on IP structures to change to. Then it might be possible to just pick a structure when there is a need for change.

4.1.3 Lack of Understanding of Parties´ Contribution

When negotiating terms for the cooperation, there is a risk that there is a lack of understanding of the contributions of each party. For example, one party´s intention with the cooperation could be to learn from each other to develop IP on their own (*Sideground Knowledge*). Meanwhile, the other party might expect the cooperation to have a project (*Foreground Knowledge*) in addition to this. It can also be a lack of understanding related to contribution in terms of capital, management, or R&D resources. Below are some examples of questions which could be beneficial to discuss:

- In terms of capital, management, and R&D resources, what is the incumbent firm expected to contribute?
- When the start-up wants to raise capital, for instance through a new share issue, should there be any milestones that the start-up must achieve? If yes, what are the milestones that have to be achieved for further financing?
- What *Background Knowledge* is each party expected to bring into the cooperation? How should the framework for *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge* be designed?

By discussing questions related to contribution expectancy, such as the three above, the risk of conflicts or uncertainty decreases. It is beneficial to raise this topic in an early stage of the *Deal Structuring* to create clarity and strengthen the cooperation.

4.1.4 Requirements on Negotiated Agreement

Depending on the complexity of the cooperation form, the requirements for the negotiated agreements vary. For instance, if there is an IP structure which contains several variants of *Foreground Knowledge* (i.e. innovative work made together within a project or the start-up´s R&D) the agreement needs to cover the IP management related to all of this. Additionally, the management needed for *Background Knowledge*, *Sideground Knowledge*, and *Postground Knowledge* impact the requirements on the agreement.

4.2 Key Intellectual Property Challenges Derived From Semi-Structured Interviews

For analysis of the data collected, a data structure was created with inspiration from D. A. Gioia et al.[31]. The building of the data structure could be described as the pivotal step of the entire research approach. Moreover, the data structure is grounded in the *1st Order Concepts*, *2nd Order Themes*, and *Aggregate Dimensions*. 1st order analysis uses informant-centric terms and codes, meanwhile, 2nd order analysis uses researcher-centric concepts, themes, and dimensions. *Aggregate Dimensions* are directly based on *2nd Order Themes* and can be interpreted as the main results of the Gioia data structure. The data structure not only allowed configuration of the data into a meaningful visual aid but also provided a graphic representation of the progression from data to terms and themes. This is an established and commonly used approach to demonstrate rigour in qualitative research.[31]

The *1st Order Concepts* can be interpreted as key quotes and statements from respondents in the semi-structured interviews for the data structure created. Based on these, IP challenges were identified and presented as *2nd Order Themes*. From these themes, key IP challenges were identified and presented as *Aggregate Dimensions*. These were as follows: *Legal and operational vulnerabilities*, *Divergent IP perceptions*, and *Process and governance gaps*. In Figure B.1, the data structure (i.e. the progression from *1st Order Concepts* to *Aggregate Dimensions*) based on data collected in the semi-structured interviews and inspired by D. A. Gioia et al.[31] is presented.

4.2.1 Legal and Operational Vulnerabilities

Patent quality and coverage are both important to consider to avoid *Legal and operational vulnerabilities*. Moreover, all parties involved in the cooperation must have their needs satisfied in some way. This means that, for instance, the coverage of an IP is aligned with the strategic vision of both the incumbent firm and the start-up. Respondent H reflects on the criteria needed to be satisfied for an incumbent firm to be willing to invest in a start-up:

"For you [the incumbent firm] to be able to invest, I think it is necessary to know that the patents protect what we [the incumbent firm and start-up] want to do together. It could be that it protects what the start-up wants to do, but not the incumbent firm´s interests."

- Respondent H: Senior IP & Technology Advisor

In addition to patent quality and coverage, contamination risks are important to consider in the *Legal and operational vulnerabilities* context. IP contamination is the involuntary transfer of IP between parties. Respondent B describes IP contamination as a challenge with harmful consequences that is preferably avoided. This respondent also suggests approaches that are effective in preventing IP contamination:

> *"It would be preferable if there were patent applications where the other party defines their unique solution and how it works. This would prevent infection with confidential information."*
>
> - Respondent B: Director, IP Support

It is also of importance to consider open source and licensing risks to avoid *Legal and operational vulnerabilities*. Open source software has transformed how technology is developed and used. It facilitates cooperation that allows developers to access, modify, and distribute code. Often, there is no need to pay licence fees to an upstream licensor. However, the open source model could clash with the traditional IP practices that aim to secure a monopoly for innovators. Balancing the benefits of cooperation with the need to protect proprietary technology is a complex undertaking that requires a nuanced approach. It can be highly complex to balance the need for proprietary technology protection with the benefits of open source cooperation.[32] Respondent C explains the importance of being careful when dealing with open source software:

> *"You [the incumbent firm] do not want majority ownership in a start-up if it infects the parent company [the incumbent firm] with critical open source software. The biggest fear is to be associated with a start-up that uses open source software that gives other free access to all of its patents. This could cause the others to claim the rights of the parent company´s [the incumbent firm´s] patents, which you do not want."*
>
> - Respondent C: Director, Patent Unit

One example of where licensing risks could occur is in brand licensing. This is defined as the process of leasing a trademarked or copyrighted property for use in connection with a product, service, or promotion.[33] Respondent M reflects on the challenge with start-ups that use incumbent firms´ brands without permission, explains how this could be dealt with, and presents some possibilities involved in brand licensing in corporate venturing:

"Several start-ups try to use another firm´s brand without authorisation. This is handled with enforcement actions. [...] Brand licensing could be interesting to adapt in corporate venturing, for instance, if we [the incumbent firm] want to invest in a certain technology [the start-up] that has a product they want to put on the market. Here, it is possible to find a solution where they [the start-up] receive the permission to use our [the incumbent firm´s] brand to develop their business."

- Respondent M: VP, Head of IP

In Figure 4.6, the first part of the data structure is presented (i.e. the progression from *1st Order Concepts* to *Aggregate Dimensions* for the key IP challenge *Legal and operational vulnerabilities*).

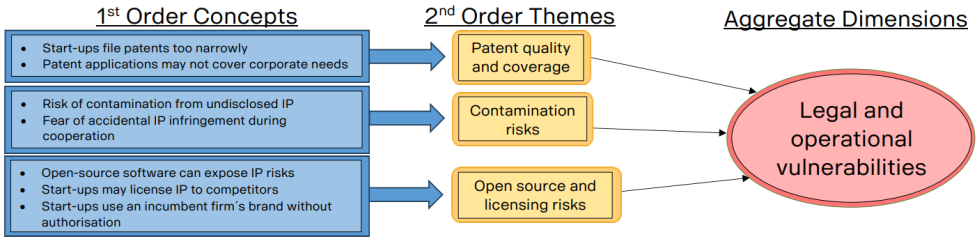


Figure 4.6: First part of data structure based on data collected in semi-structured interviews and inspired by D. A. Gioia et al.[31]

4.2.2 Divergent Intellectual Property Perceptions

In the context of *Divergent IP Perceptions*, lack of IP knowledge in start-ups and asymmetry in IP valuation are two common problems. For instance, the start-ups tend to overvalue their IP. Respondent H explains the lack of IP knowledge that start-ups could have:

"They [the start-ups] can work with known technology, file a patent application, and manage to have it approved. However, they overvalue their IP since there is a lack of understanding of the number of patents out there and who has the know-how and IP knowledge."

- Respondent H: Senior IP & Technology Advisor

Another challenge that could appear, in the context of *Divergent IP perceptions*, is exclusivity conflicts. Here, there are different opinions between the incumbent firm and start-up. The incumbent firm wants exclusive

treatment by the start-up (exclusive rights to IP, etc.), meanwhile, the start-up wants to avoid exclusivity in favour of cooperation with other clients. Respondent N discusses the challenge of exclusivity conflicts and suggests a potential solution to this by giving an example:

"The incumbent firm wants to have control and exclusivity. The start-up, however, wants to give the incumbent firm some sort of non-exclusive license that is royalty-bearing. This is a potential future revenue stream for the start-up and opens up for cooperation with other clients. [...] You [the incumbent firm and start-up] can try to find a license structure that considers field-of-use, which means having exclusivity in a certain area. For instance, we [the incumbent firm] want to have exclusivity towards all our competitors in the vehicle industry, but you [the start-up] are free to give out licenses intended for use in another industry."

- Respondent N: Patent Manager

In Figure 4.7, the second part of the data structure is presented (i.e. the progression from *1st Order Concepts* to *Aggregate Dimensions* for the key IP challenge *Divergent IP perceptions*).

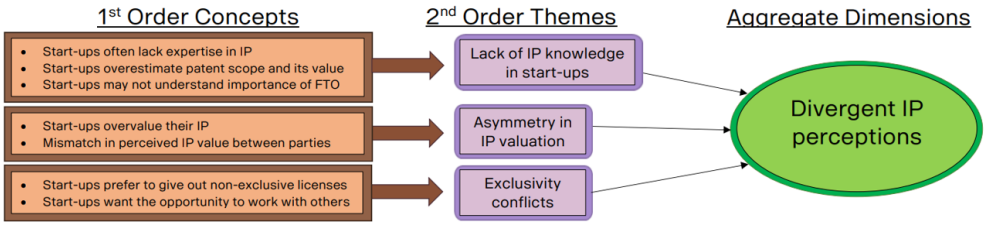


Figure 4.7: Second part of data structure based on data collection from semi-structured interviews and inspired by D. A. Gioia et al.[31]

4.2.3 Process and Governance Gaps

A challenge, which goes under *Process and governance gaps*, is contractual gaps. The contractual part is critical since the management of *IP Rights Assignment* highly impacts the outcome of the cooperation between incumbent firm and a start-up. Respondent B comments on this:

"If you [the incumbent firm and start-up], in an early stage, can agree on how to distribute the rights to the IP that is most preferable. Together, develop an agreement which covers as many IP challenges as possible."

- Respondent B: Director, IP Support

Another aspect to consider in the contractual part is the type of cooperation to be applied. Furthermore, there are both pros and cons to different cooperation forms. For instance, joint ownership of IP can be one way of managing IP rights. Respondent M presents one challenge related to joint *IP Ownership*:

> *"Joint ownership of IP is something that we [the incumbent firm] avoid because our experience is that it can be hard. [...] In some cases where we have had joint ownership of IP, the other partner has not taken any responsibility for their part of the annual fee payment, meaning that we had to cover their payment."*
>
> - Respondent M: VP, Head of IP

In addition to contractual gaps, one challenge that can be faced is cultural and operational misalignment. For example, start-ups require high speed in the processes and tend to manage trade secrets poorly compared to incumbent firms. Respondent C discusses the concern of bad trade secret management in cooperation between incumbent firm and start-up:

> *"For the incumbent firm, there are clear routines for protection of trade secrets. However, the start-up is super fast in its processes and most often has no routines for this."*
>
> - Respondent C: Director, Patent Unit

In Figure 4.8, the third part of the data structure is presented (i.e. the progression from *1st Order Concepts* to *Aggregate Dimensions* for the key IP challenge *Process and governance gaps*).

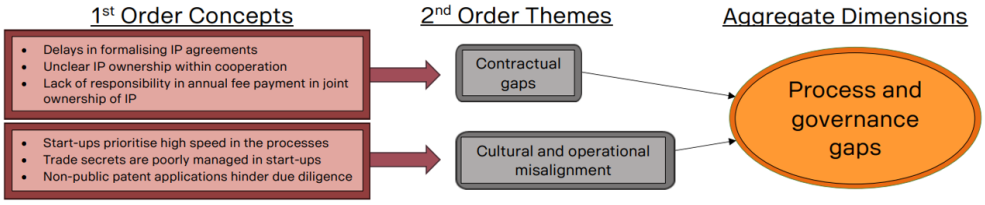


Figure 4.8: Third part of data structure based on data collection from semi-structured interviews and inspired by D. A. Gioia et al.[31]

4.3 Relation Between Key Intellectual Property Challenges and Venture Capital Investment Process

The key IP challenges derived from the IP structures and the semi-structured interviews are compiled in Figure 4.9.

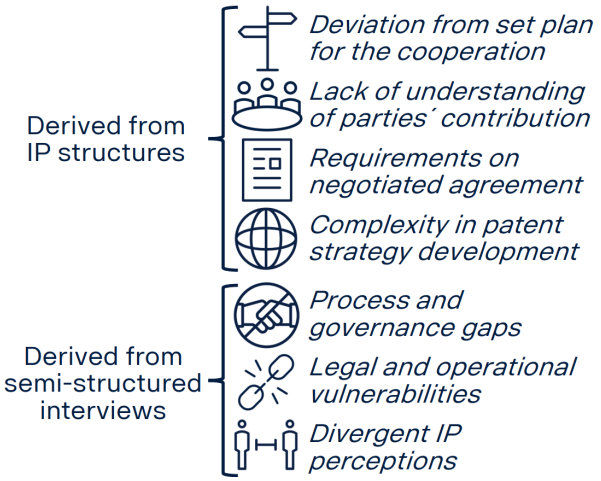


Figure 4.9: The key IP challenges derived from the IP structures and the semi-structured interviews.

The key IP challenges are related to different steps of the VC investment process. Figure 4.10 shows the relation between the key IP challenges and the VC investment process.

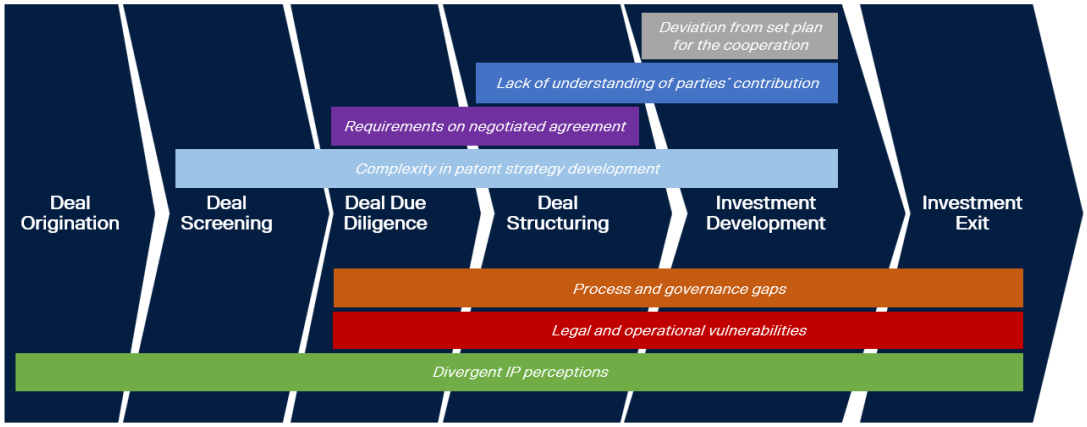


Figure 4.10: Relation between key IP challenges and VC investment process.

From subsection **4.3.1 Key Intellectual Property Challenges in Deal Origination** to subsection **4.3.6 Key Intellectual Property Challenges in Investment Exit**, practical guidelines for fruitful management of key IP challenges are provided for each step of the VC investment process. Moreover, this follows the relationship between the key IP challenges and the VC investment process presented in Figure 4.10.

4.3.1 Key Intellectual Property Challenges in Deal Origination

Divergent IP perceptions: In this first step of the VC investment process, the goal is to obtain an adequate deal flow that is high in both quality and quantity. When sourcing the deal flow, it is beneficial to pay attention to how the start-up understands the value of its IP. This is one step in ensuring that its IP is aligned with the incumbent firm´s strategic goals. To obtain an idea of the start-up´s IP valuation, one could perform an IP landscape as well as an assessment of the start-up´s patent portfolios.

4.3.2 Key Intellectual Property Challenges in Deal Screening

Divergent IP perceptions: When selection of the most promising investment opportunities is made, it is important that the incumbent firm analyses geographic coverage and conducts a know-how assessment. This type of assessment contains an investigation of the start-up´s IP, team, market capabilities, position compared to competitors, and operational & executional capabilities. When investigating IP, it is important to do FTO and evaluate if the start-up has FTO analysis routines. Performing FTO is necessary since the risk of IP infringement is otherwise high. By finding start-ups which satisfy the criteria connected to this investigation, divergent IP perceptions are avoided.

Complexity in patent strategy development: Through the know-how assessment, the incumbent firm can create an idea of the start-ups´ patent strategy approach. Moreover, studying the IP of the start-ups could give a hint of their expectations of the incumbent firm in the development of a joint patent strategy. Interesting to know is, for example, whether the start-up has trade secrets or not and the geographic as well as patent coverage.

4.3.3 Key Intellectual Property Challenges in Deal Due Diligence

Divergent IP perceptions: Here, further investigation is made with the results from the *Deal Screening* as a basis. Investigating IP, i.e. patents, trade secrets, ownership, licensing agreements, etc., in more detail is essential in the *Deal Due Diligence*.

Complexity in patent strategy development: As in the *Deal Screening*, the *Deal Due Diligence* allows investigation that provides an understanding of the start-ups´ view on patent strategy.

Requirements on negotiated agreements: Through due diligence, it is possible to find characteristics of the start-up which add requirements on the agreement in case cooperation is initiated. From the beginning, the incumbent firm has an idea of the start-up, meaning they already from start reflect on the layout for an agreement. However, discoveries about the start-up could change the incumbent firm´s expectation of possible outcomes from negotiation, meaning that new requirements could arise.

Legal and operational vulnerabilities: One possibility for the incumbent firm, when performing *Deal Due Diligence*, is to investigate the start-ups´ use of open source software. If open source software is used, IP risks could be exposed. It is therefore good to know about this in an early stage, to prepare for managing it.

Process and governance gaps: Here, the risk of start-ups having non-public patent applications should be considered. This hinders *Deal Due Diligence* since the incumbent firm is unable to find non-public patent applications. Moreover, this means the incumbent firm cannot obtain an understanding of the total patent portfolio.

4.3.4 Key Intellectual Property Challenges in Deal Structuring

Divergent IP perceptions: When negotiating, there is a risk that different opinions about certain aspects arise. To have a chance to solve this, both the incumbent firm and start-up have to be willing to make compromises.

Complexity in patent strategy development: The results of the *Deal Structuring*, i.e. the negotiated agreement, highly affect the development of a suitable patent strategy. This is good to be aware of, since it is necessary to be able to develop a patent strategy which satisfies the goals and visions.

Requirements on negotiated agreements: Depending on the IP structure chosen as well as other criteria set for the cooperation, the requirements put on the negotiated agreements vary. Moreover, being aware of this and willing to manage all the clauses needed for the agreement is important. If this is not achieved, the agreement will not be aligned with the strategic goals of the cooperation between the incumbent firm and start-up.

Legal and operational vulnerabilities: It is important to do risk management in the *Deal Structuring*, such as treating risks related to IP contamination, open source software, and licensing. Furthermore, to avoid accidental IP infringement and contamination from undisclosed IP, the negotiation needs to be transparent.

Process and governance gaps: To avoid unclear *IP Rights Assignment*, it is necessary to develop a plan which covers IP structures possible to use as representation for the cooperation form. To do this as early as possible in the *Deal Structuring* is preferable, since delays in formalising IP agreements tend to cause uncertainty.

Lack of understanding of parties´ contribution: The *Deal Structuring* is very critical for the management of contributions from each party. Here, guidelines and frameworks could be developed to outline the expectations for both the incumbent firm and start-up. To manage *Background Knowledge*, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*, regulations for additional funding from the incumbent firm to start-up, and support in the form of management and R&D resources are crucial. Otherwise, there will be a lack of understanding of each other´s contributions.

4.3.5 Key Intellectual Property Challenges in Investment Development

Divergent IP perceptions: Throughout the *Investment Development*, i.e. when the cooperation is ongoing, project work may take another turn. One of the parties might find an issue with the cooperation form, meaning that there is an interest in changing it. Here, it is beneficial to be flexible and able to accept changes to keep the cooperation moving forward.

Complexity in patent strategy development: When managing the IP produced during the *Investment Development*, the incumbent firm and start-up need to agree on the approach for disclosing and protecting the IP. For example, there has to be alignment on how and when inventions are disclosed to avoid anticipating patent applications. The parties should also discuss and agree on how to protect different types of IP, i.e. guidelines for when to file for patents and when to keep secret.

Legal and operational vulnerabilities: To respect the interest of each party, with the negotiated agreement as a basis, is important. This covers, for instance, treatment of *IP Rights Assignment* and *IP Licensing*. If the agreement, for instance, states that the incumbent firm should be assigned the rights for all *Foreground Knowledge* produced within the *Investment Development*, this has to be followed. Another example could be if the agreement states that a license structure which considers field-of-use should be applied. This is important to follow, since this is about having exclusivity towards competitors in a certain area.

Process and governance gaps: It is appropriate to set up processes for regular monitoring of the agreement and how the parties handle their respective obligations. In case any of the parties does not fulfil their commitment, it is appropriate to have measures prepared. For example, failed commitments could be a lack of responsibility in annual fee payment where joint ownership of IP is applied, or mismanagement of trade secrets. Here, preventive measures need to be taken. In the agreement, a clause which gives the right to terminate if the other party fails to meet the key obligations could be involved. To have open and transparent communication, meaning there is a dialogue between the parties to address concerns before they escalate, is beneficial. Then, it is important that the parties agree on the meaning of the terms involved in the agreement, where there is room for interpretation, and why that is the case.

Lack of understanding of parties´ contribution: Throughout the *Investment Development*, there is a risk that a lack of understanding of the parties´ contribution arises. To deal with this, and avoid further concerns and harm to the cooperation, communication and willingness to compromise are key.

Deviation from set plan for the cooperation: If there is any deviation from the set plan, the IP related to this deviation need to be managed properly. Preferably, there is a management plan for this prepared and included in the contract. This means that there is an agreement on contingency plans to activate if needed.

4.3.6 Key Intellectual Property Challenges in Investment Exit

Divergent IP perceptions: In the process of preparing an *Investment Exit*, there are several agreements which have to be checked and finalised. The management of *Postground Knowledge* is essential, since it is likely that this IP includes improvements made on *Background Knowledge*, *Foreground Knowledge*, and *Sideground Knowledge*.

Legal and operational vulnerabilities: When planning for *Investment Exit*, to ensure everyone is on the same track is critical. To avoid start-ups licensing IP to competitors or using the incumbent firm´s brand without authorisation, one way is to create clarity before *Investment Exit*, as well as following up on each other´s work post *Investment Exit*.

Process and governance gaps: As mentioned previously, management of *Postground Knowledge* is essential. When reaching *Investment Exit*, it is necessary to have this under control to prevent confusion connected to each party´s right of ownership and licensing.

Chapter 5

Discussion

In this project, the following research questions were investigated:

1. What key challenges are encountered when managing IP in corporate-backed ventures?
2. What practical guidelines are needed for fruitful management of key IP challenges in corporate-backed ventures?

The first research question is addressed by the key IP challenges derived in **Chapter 4. Derivation of Key Intellectual Property Challenges**: *Complexity in patent strategy development*, *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Legal and operational vulnerabilities*, *Divergent IP perceptions*, and *Process and governance gaps*. These were the key IP challenges identified, covering *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage* discussed in **Chapter 2. Analytical Framework**. Through research which covers IP structures, uses literature on IP challenges, and relies on data collected from interviews with experts, the key IP challenges were identified and linked to the VC investment process.

By studying the key IP challenges, practical guidelines for fruitful management of IP challenges in corporate-backed ventures were developed, thereby addressing the second research question. Practical guidelines for each step of the VC investment process are found in **Chapter 4. Derivation of Key Intellectual Property Challenges** and are compiled in the Global IP Playbook presented in Appendix C. The playbook is designed to support Scania VNB in strengthening its IP management and mitigating IP risks in its investments. By outlining relevant IP challenges and providing practical tools, the playbook

enhances alignment between Scania IP and Scania VNB. It contributes to a shared understanding of IP challenges in corporate venturing and enables more proactive, coordinated support when evaluating and engaging with external innovation partners. The Global IP Playbook is a first version which is intended to develop over time, incorporating new insights and experiences as it is applied in practice.

5.1 Relation Between Previous Knowledge and Findings

IP Strength, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*, as well as the IP types (i.e. *Background Knowledge*, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*), were connected with the VC investment process in **Chapter 2. Analytical Framework**. From this previous knowledge, further research was conducted.

5.1.1 Change in Cooperation Form and Impact on Outcome of the Cooperation

The concept of IP structures was further developed to identify several cooperation forms in corporate venturing. By continuing from where previous knowledge left off, i.e. the IP structure inspired by O. Granstrand et al.[11] (see Figure 2.2) and the IP structure also inspired by an interview with M. Holgersson (see Figure 2.4), the IP structures presented in Figures 4.1-4.5 were developed. Figure 2.4 and Figures 4.1-4.5 have in common that they are all cooperation forms which fit in the context of corporate venturing, since they all apply in the VC investment process. The IP structures in Figures 4.1-4.5 added to previous knowledge in the sense that they raised the idea that there are several cooperation forms to choose between, not only one, for cooperation between an incumbent firm and a start-up. These findings showed that changes in the cooperation form are possible and that these changes impact the outcome of the cooperation. *Foreground Knowledge* can be produced in a project or within the start-up´s business. It is possible for each party to produce *Sideground knowledge*, except when there is *Foreground Knowledge* arising from the start-up´s business. Here, the incumbent firm is the only one enabled to produce *Sideground knowledge*. To choose a cooperation form that enables involved parties to produce the type of IP wanted during the *Investment Development* is crucial. To have contingency plans prepared and ready to execute when a change in cooperation form is needed would benefit the cooperation.

5.1.2 Transparency, Risk Management, and Regular Monitoring Within the Cooperation

In the context of cooperation between an incumbent firm and a start-up, IP challenges and management of these were investigated by conducting semi-structured interviews with experts. The knowledge gathered from the literature review was used as a basis when conducting these interviews. Here, the IP types described by O. Granstrand et al.[11] and the general IP challenges acted as a basis. The interviews gave data, which validated previous knowledge but also added new one. For instance, the data validated previous knowledge by addressing the importance of performing know-how assessment and FTO. Then, the data added to previous knowledge by, for instance, addressing the risk of conflicts related to lacking responsibility in annual fee payment in joint ownership of IP and start-ups´ use of incumbent firms´ brands without authorisation. In total, the data highlighted the importance of being transparent, conducting risk management, and having regular monitoring of the agreement within the cooperation.

5.1.3 Identification of Seven Key Intellectual Property Challenges

The *Complexity in patent strategy development*, *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Legal and operational vulnerabilities*, *Divergent IP perceptions*, and *Process and governance gaps* are the seven key IP challenges identified in this master thesis project. These are briefly summarised below:

- *Complexity in patent strategy development* is related to patent coverability and approach for protecting IP. For example, whether to keep the invention a trade secret or to obtain patent protection. The incumbent firm might prefer to keep the invention a trade secret because it sees a potential risk that competitors could easily circumvent the protection or that it is hard to prove IP infringement, meanwhile, the start-up might want patent protection to obtain revenue streams through licensing.

- *Deviation from set plan for the cooperation* is related to a change in the cooperation form, which is executed because the original plan no longer meets the needs of the parties involved. Having contingency

plans prepared for this change is beneficial, as it facilitates the change in cooperation form.

- *Lack of understanding of parties´ contribution* is, for instance, about each party´s contribution in terms of capital, management, R&D resources, test facilities, user data, and IP. Here, it may be of interest to establish milestones for the start-up to achieve to receive further funding (such as a new share issue, for example). To develop a plan for the cooperation, which covers these, is crucial.

- *Requirements on negotiated agreement* depends on the cooperation form, where *Foreground Knowledge* within the cooperation puts pressure on the requirements since the *IP Rights Assignment* could be complicated to manage. Here, where the incumbent firm and start-up produce IP together, it is preferable if the agreement contains guidelines for managing the IP.

- *Legal and operational vulnerabilities* covers challenges related to patent quality, IP contamination, and open-source software. These are all important to consider since they are critical and highly impact the outcome of the cooperation.

- *Divergent IP perceptions* is about potential differences in IP knowledge, valuation, and view on exclusivity. For instance, the start-up might prefer to give the incumbent firm a non-exclusive license to enable cooperation with other incumbent firms.

- *Process and governance gaps* covers challenges such as cultural and operational misalignment as well as contractual gaps. For instance, it is possible to have misalignment in the management of trade secrets within the corporation.

When comparing the key IP challenges with previous knowledge, it is found that these cover the general IP challenges *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*. However, they are adapted to corporate-backed ventures to also cover those challenges that are unique to this context. In Table 5.1, examples of validation (green text) of previous knowledge (**Chapter 2. Analytical Framework**) and examples of new findings (red text) found in results (see **Chapter 4. Derivation of Key Intellectual Property Challenges**) in the context of *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage* are found.

Table 5.1: Examples of validation (green text) of previous knowledge (**Chapter 2. Analytical Framework**) and examples of new findings (red text) found in results (see **Chapter 4. Derivation of Key Intellectual Property Challenges**) in the context of *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*.

	Chapter 2. Analytical Framework	**Chapter 4. Derivation of Key Intellectual Property Challenges**
IP Strength	Covered in subsection 2.1.3: • IP landscape analysis to evaluate IP terrain in a certain tech sector, industry, or market • Know-how assessment and FTO analysis	Covered by *Complexity in patent strategy development*: • There could be different opinions about patent coverability, i.e. which markets to obtain IP protection in and what technology to protect • One potential challenge is to agree on which way to protect the IP (e.g. patent or trade secret) Covered by *Divergent IP perceptions*: • Start-ups may lack understanding of who has the Know-how and not understand the importance of FTO
IP Ownership	Covered in subsection 2.1.3: • Due diligence to ensure start-ups have a clear title to their IP and no third-party ownership claims	Covered by *Process and governance gaps*: • Non-public patent applications hinder due diligence • Lack of responsibility in annual fee payment in joint ownership of IP
IP Rights Assignment	Covered in subsection 2.1.3: • Regulate rights assignment of IP arising within cooperation	Covered by *Process and gouvernance gaps*: • Most preferable if the parties agree on how to distribute the rights to the IP
IP Licensing	Covered in subsection 2.1.3: • Start-ups promote scalable license agreements	Covered by *Divergent IP perceptions*: • Start-ups want the opportunity to work with others and therefore prefer to give out non-exclusive licenses Covered by *Legal and operational vulnerabilities*: • Start-ups use incumbent firm´s brand without authorisation
IP Leakage	Covered in subsection "2.1.3": • Safeguarding of trade secret information • Ensure that start-ups appropriately store, monitor, and restrict access to source code as well as other valuable data (use NDAs)	Covered by *Process and gouvernance gaps*: • Poor management of trade secrets in start-ups Covered by *Legal and operational vulnerabilities*: • Open source software can expose IP risks • Risk of IP contamination from undisclosed IP • Risk of accidental IP infringement

5.2 Limitations

In the context of corporate venturing, there are several parties involved. For instance, a start-up could have several CVC firms as investors as well as a university connection. This means that there are several worlds to consider when managing cooperation of this kind. In this master thesis project, interviews were conducted in such a way that respondents´ backgrounds were related to academic institutions, incumbent firms, and start-ups. Moreover, the study is based on the incumbent firm´s perspective.

Collaboration with academia differs from collaborations with companies in some important respects. For example, there could be special legislation around the researchers' rights to the inventions, and public funding could play a different role. University-related challenges were partly captured in this project by including one participant from a Swedish university as well as one participant from a Swedish university incubator, but future studies could make important contributions to this work by exploring that setting more thoroughly. For example, Technology Transfer (TT) at universities such as Cambridge in the UK or Harvard in the USA could be investigated. This might highlight eventual differences in TT between countries.

Chapter 6

Conclusion

The research made in this master thesis project highlights the need for practical guidelines to enable fruitful management of IP challenges in corporate-backed ventures. Moreover, the key IP challenges identified, i.e. *Complexity in patent strategy development*, *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Legal and operational vulnerabilities*, *Divergent IP perceptions*, and *Process and governance gaps*, show that IP is of main importance throughout the entire VC investment process and should not be underestimated. Throughout the VC investment process, which starts with *Deal Origination*, continues with *Deal Screening*, *Deal Due Diligence*, *Deal Structuring*, as well as *Investment Development*, and ends with *Investment Exit*, these key IP challenges need to be dealt with individually. It is essential to choose a cooperation form (IP structure) which satisfies all parties regarding *Background Knowledge*, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*, as well as securing good conditions for *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*. Within the cooperation between the incumbent firm and start-up, it is important to be transparent, conduct risk management, jointly develop contingency plans for changes in the cooperation form, and regularly monitor the agreement.

6.1 Future Research

In the context of key IP challenges in corporate-backed ventures, future research on the university's role in the rise and development of start-ups would contribute a lot. Since several start-ups arise in the academic landscape, their role is important to consider. Another suggestion for future research is within M&A, where the findings in this report could be expanded by exploring them in this context.

The work in this project was carried out from the perspective of the incumbent firm. meaning that both process and problems were primarily considered from the incumbent firm´s POV. It would have been of interest to include the start-up´s perspective as well, since this could have provided even more input to the Global IP Playbook. However, this was not done due to limitations in time and access to suitable start-ups.

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Appendix A

Research Discussion Guide

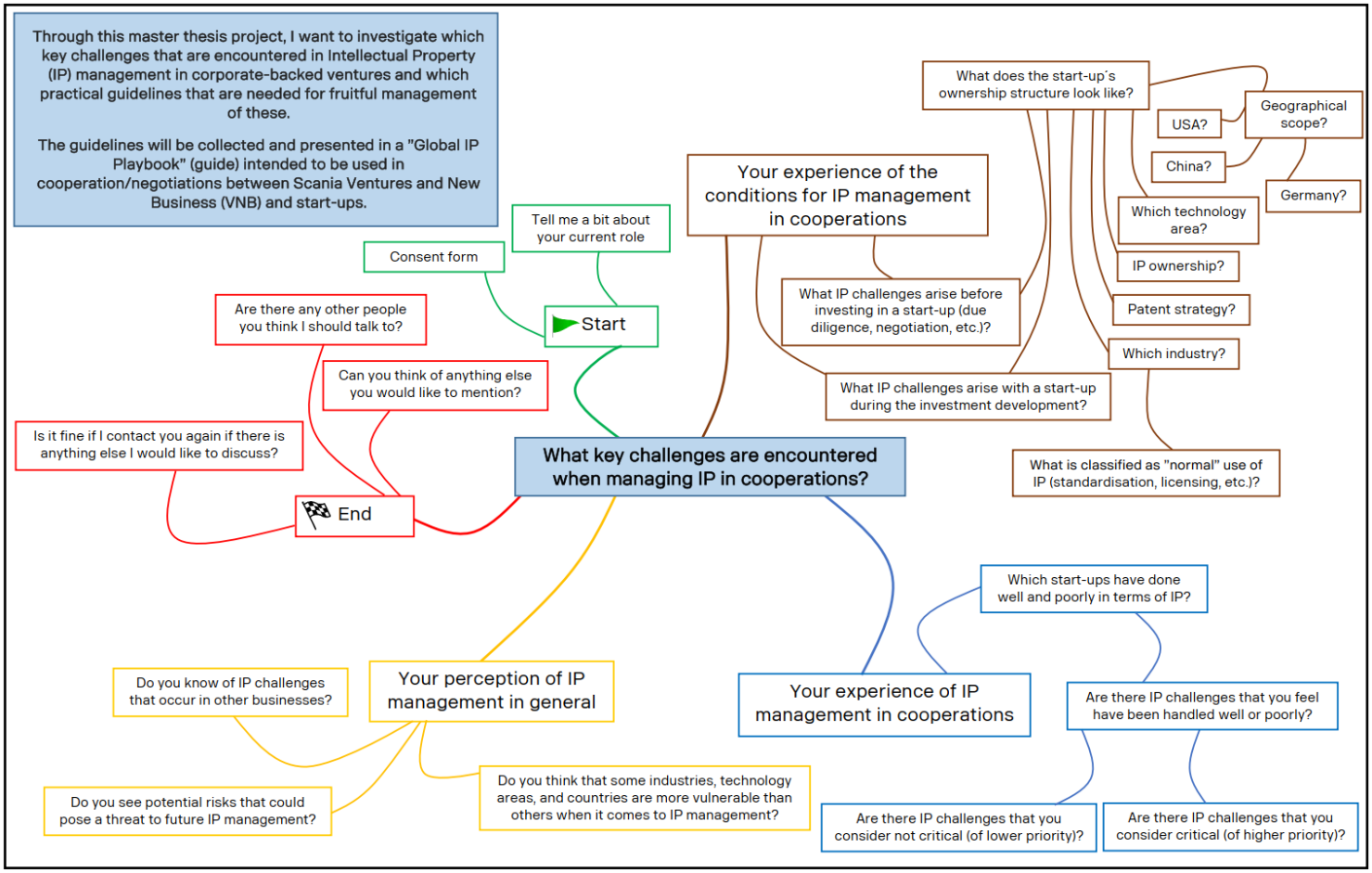


Figure A.1: Research discussion guide.

Appendix B

Gioia Data Structure

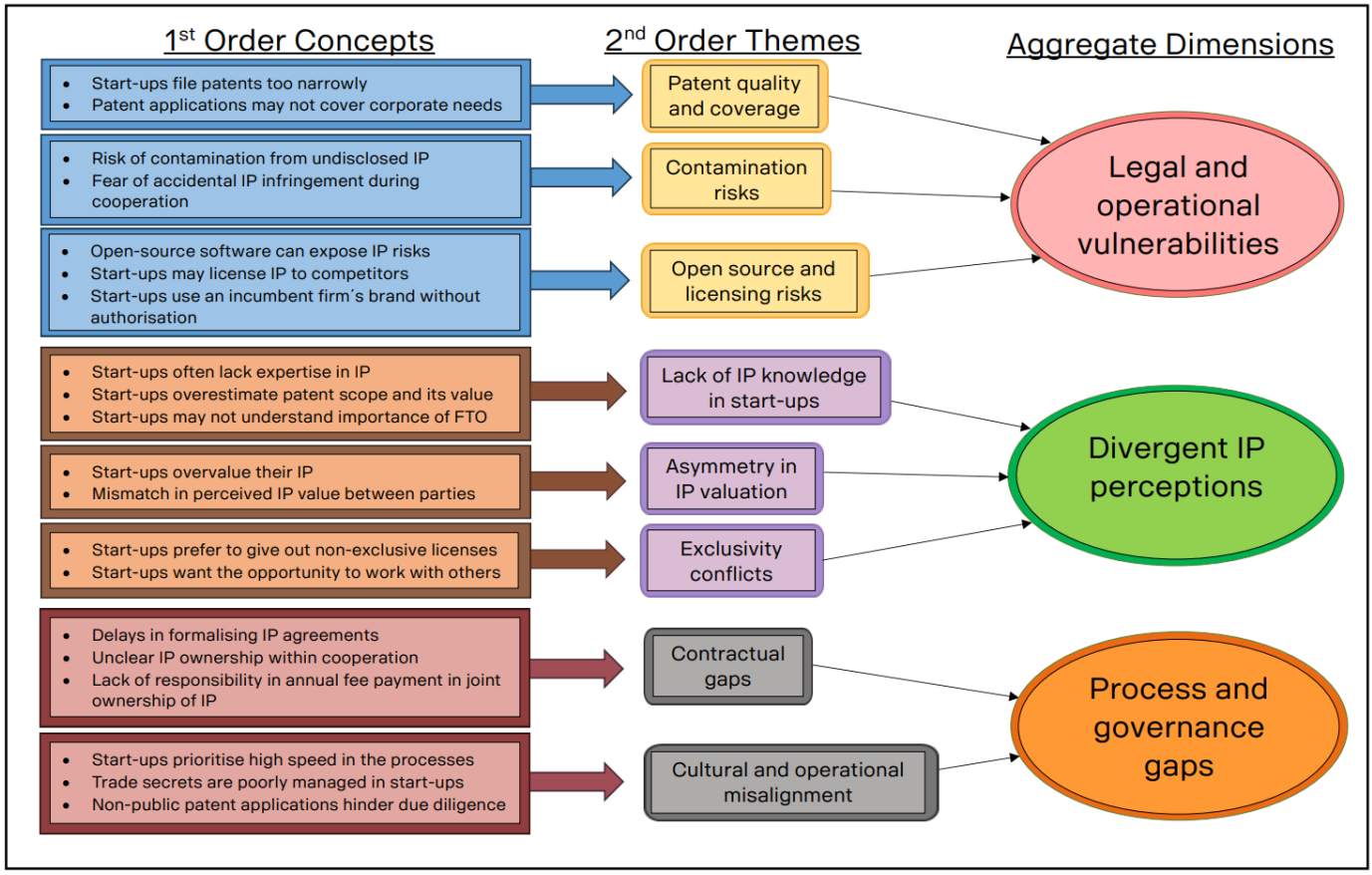


Figure B.1: Gioia data structure (i.e. the progression from *1st Order Concepts* to *Aggregate Dimensions*) based on data collected in the semi-structured interviews.

Appendix C

Global Intellectual Property Playbook



GLOBAL INTELLECTUAL PROPERTY PLAYBOOK

SCANIA INVEST
VENTURES AND NEW BUSINESS

SCANIA

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I. Management of Intellectual Property in Venture Capital

In this Global Intellectual Property (IP) Playbook, management of the key IP challenges *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Complexity in patent strategy development*, *Process and governance gaps*, *Legal and operational vulnerabilities*, and *Divergent IP perceptions* is provided. These seven key IP challenges are linked to the five steps of the Venture Capital (VC) investment process: *Source* => *Screening* => *Due Diligence* => *Execute Transaction* => *Portfolio Management*. This is presented in *Figure 1: VC investment process and its relation to key IP challenges*. The key IP challenges cover the following general IP challenges: *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage*. The Global IP Playbook enables efficient management of IP-related matters to strengthen the cooperation between Scania and Target Company (TargetCo). In this chapter, a walkthrough of the VC investment process as well as its relation to the key IP challenges and the general IP challenges is provided. To enable efficient management of these, cross-referencing is used throughout the rest of this chapter. This means it is possible to click on the challenge to be directly navigated to its detailed explanation. University-related challenges are partly captured in this work, however, future studies could make important contributions to this work by exploring that setting more thoroughly. For example, Technology Transfer (TT) at universities such as Cambridge in the UK or Harvard in the USA could be investigated. This might highlight eventual differences in TT between countries. The term *Deal Structuring* will be used throughout the playbook. This subcategory of *Execute Transaction* refers to the negotiation of legal documents for the cooperation between Scania and TargetCo.

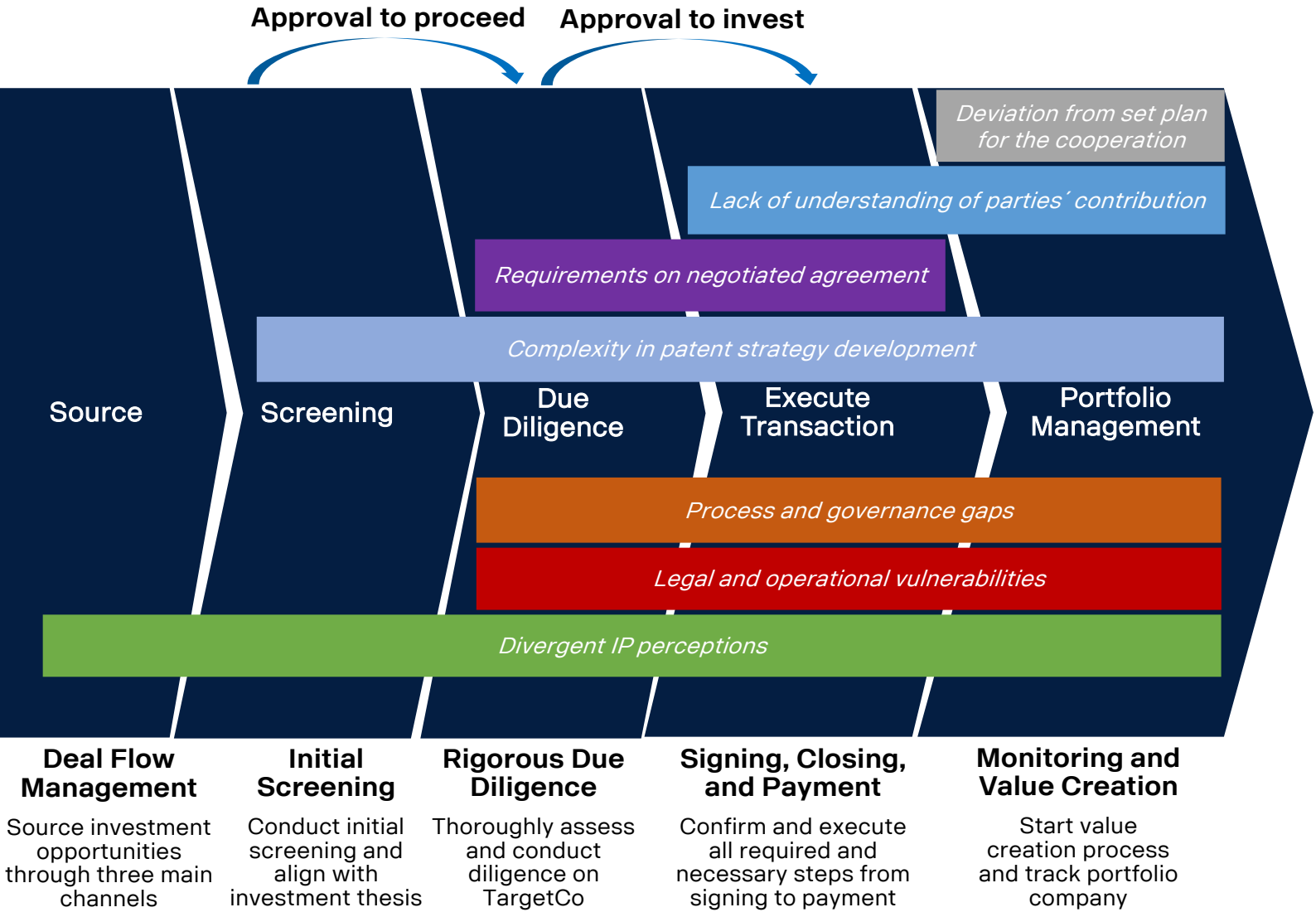
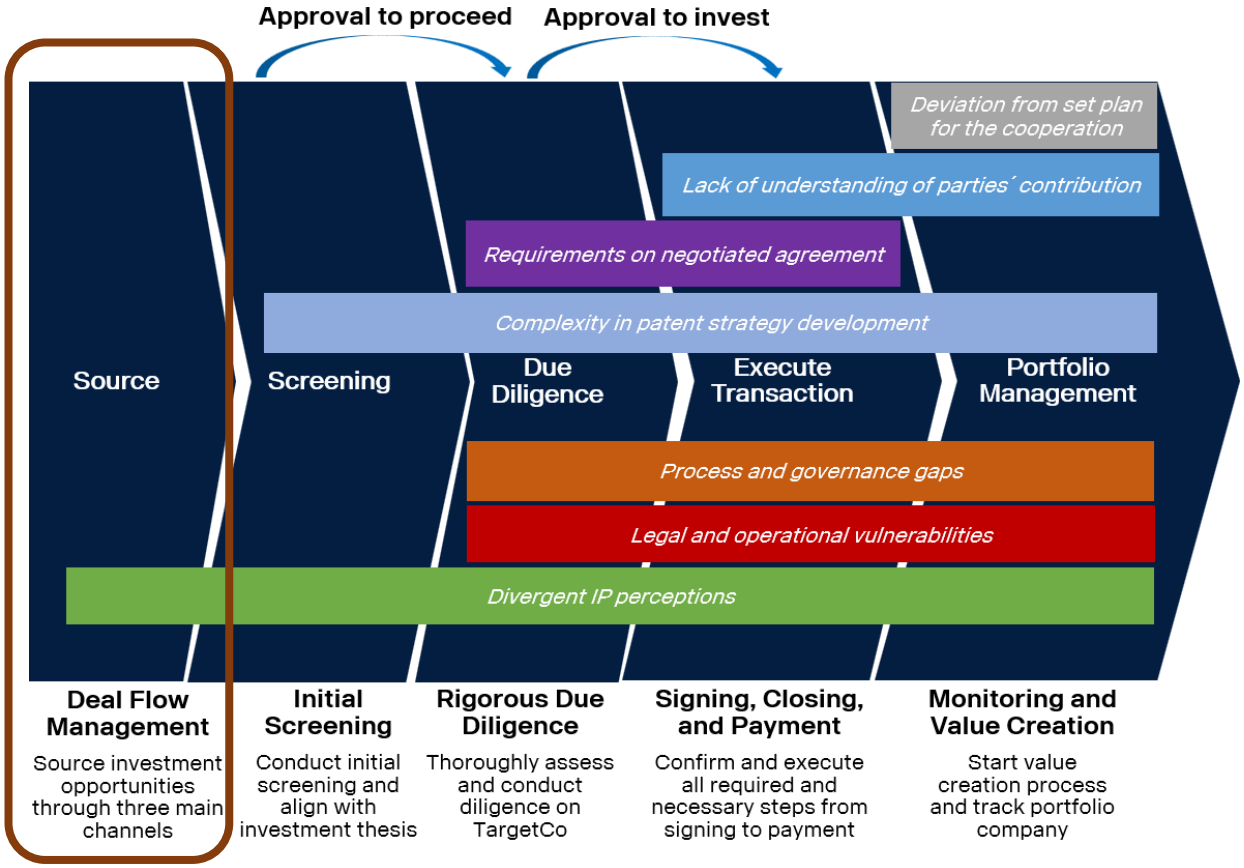


Figure 1: VC investment process and its relation to key IP challenges

1. Source



Main mission:
Secure strategic insights at the first step of the investment process.

General IP challenge:
IP Strength.

Key IP challenge:
Divergent IP perceptions.

Key checkpoints:

- Manage deal flow
- Participate in deal flow stand-ups and share lessons learned about new technologies
- Participate in meetings with key experts to obtain subject matter expertise
- Store new learnings about technologies and disruptive innovations in team

Deal Flow Management:
All deal flow is stored and managed in the Deal Flow Funnel to ensure follow-ups, tracking of facilitators, and understanding of industry exposure. The goal is to obtain an adequate deal flow that is high in both quality and quantity. When sourcing the deal flow, it is beneficial to pay attention to how TargetCo understands the value of its IP. This is one step in ensuring that its IP is aligned with Scania´s strategic goals. To obtain an idea of TargetCo´s IP valuation, one could perform an IP landscape as well as an assessment of TargetCo´s patent portfolios.

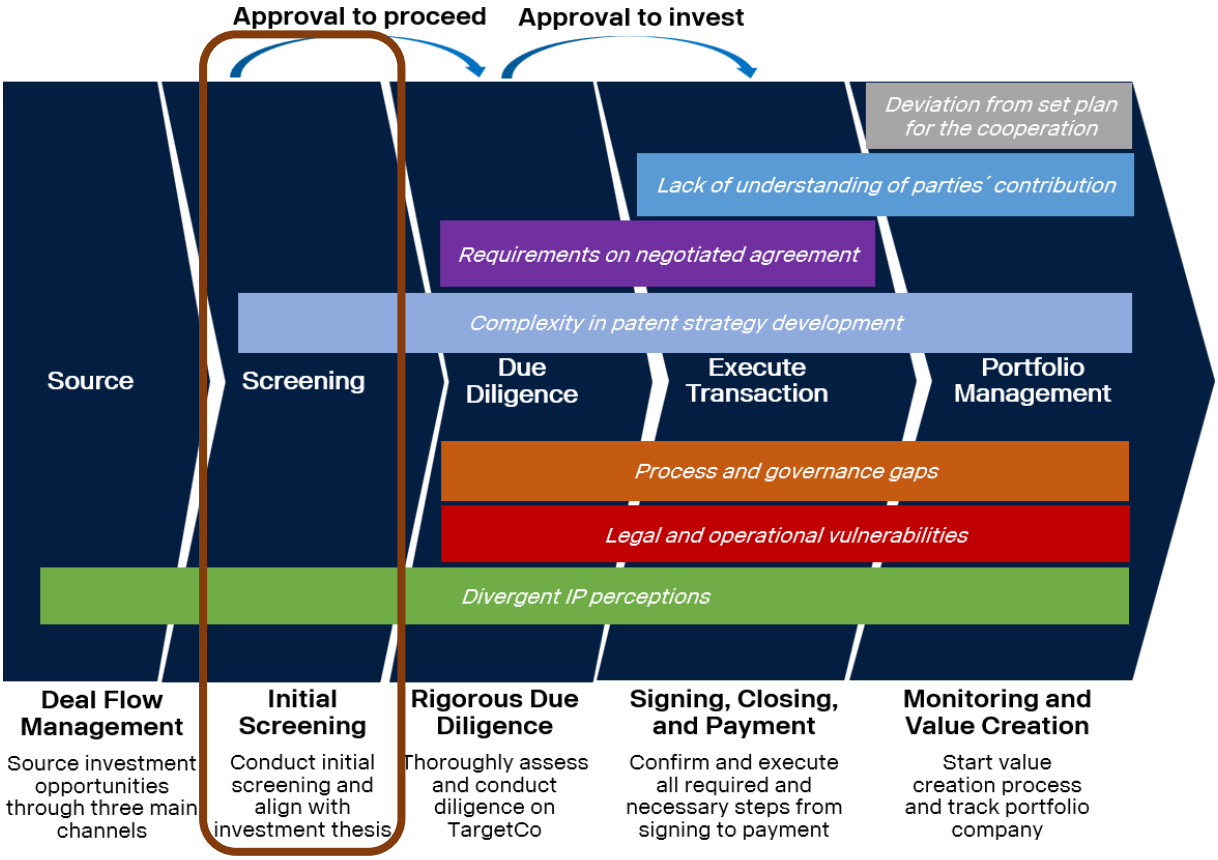
Deal Flow Stand-Ups:
Weekly deal flow stand-ups with team where weekly opportunities are presented, discussed, and lessons learned shared about new technologies. Here, it is good to share IP findings about these opportunities to see if it is possible to gain new insights and perspectives.

Subject Matter Expertise:
Bi-weekly meetings with key experts in vertical area to pulse opportunities and get a deep understanding of whether technology is strategically relevant for Scania. Here, it is beneficial to ask the key experts about their thoughts on the technology´s potential to reach the market. For instance, do they think it is possible to secure IP protection of the technology?

Self-Sharpening Tool:
New learnings about technologies and disruptive innovations are stored and shared in team, to be able to make quicker investment decisions in the future. From an IP Point Of View (POV), it is of interest to store learnings about, for instance, TargetCo's valuation of its IP. Is there a pattern where TargetCo's understanding of their IP depends on the market, technology sector, and industry they are active in?



2. Screening



Main mission:
Assess the opportunity to obtain an understanding of and a belief in it.

General IP challenges:
IP Strength, IP Ownership, and *IP Leakage.*

Key IP challenges:
Divergent IP perceptions and *Complexity in patent strategy development*.

Key checkpoints:

- Signed Scania NDA
- Access target data room
- Conduct anti-trust review/merger clearance
- Notify transaction team of upcoming potential investment
- Prepare Investment Committee (IC) material "Approval to Invest"

The selection of the most promising investment opportunities is one of the key tasks in the deal screening. Here, several screens are applied to secure certain aspects of investment strategy and efficiently reduce the number of investment opportunities. The aim of this is to receive a reduced list of deal flow that is worth further investigation. The screens applied are the following: *Industry overview*, *Financial overview*, *Management team and stakeholders*, *Technology*, *Investment rationale*, *Initial deal terms reflections*, *Scania value-add and strategic return*, and *Exit strategy and financial return*. If the criteria in these screens are not met by an investment opportunity, it is eliminated from the list. These are explained below:

Company Overview:

Industry overview – How big is it? Is it experiencing change? Is disruption possible?

Financial overview – What is the financial outlook of TargetCo, cost structure, and revenue drivers?

Management team and stakeholders – Who is the founding and supporting team? Who are the current investors?

Technology – How does the technology work and what is the level of disruption and strategic relevance?

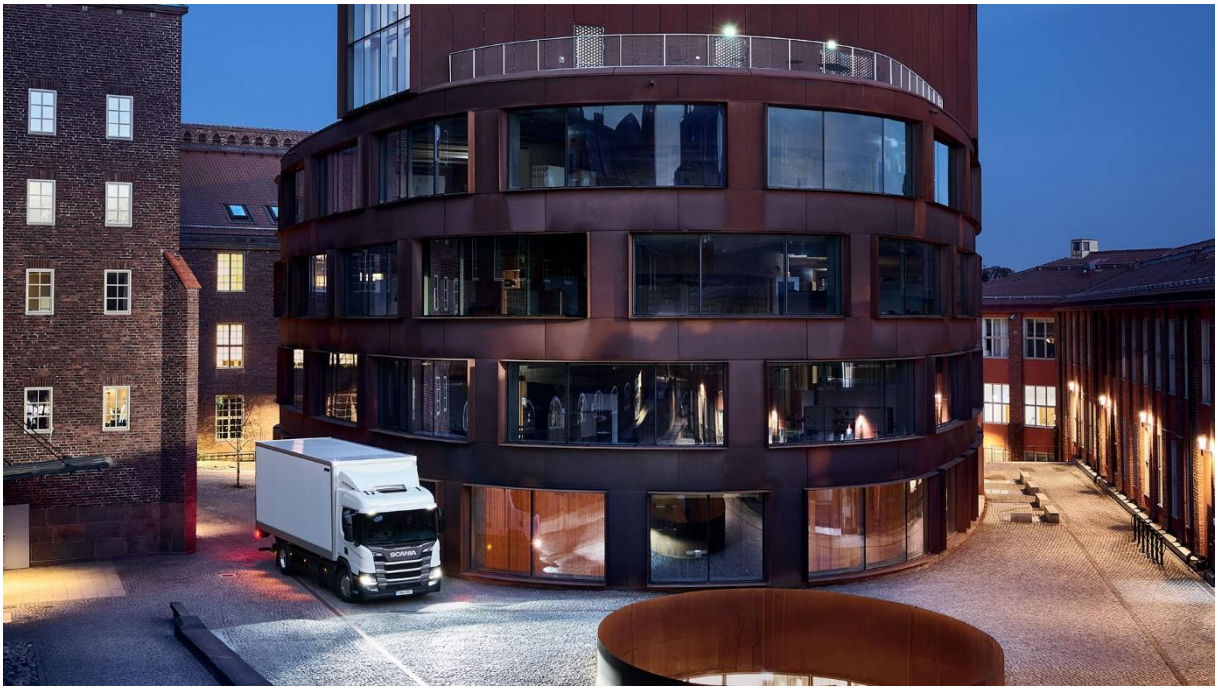
Deal Summary:

Investment rationale – Why does it make sense that Scania should invest in this TargetCo?

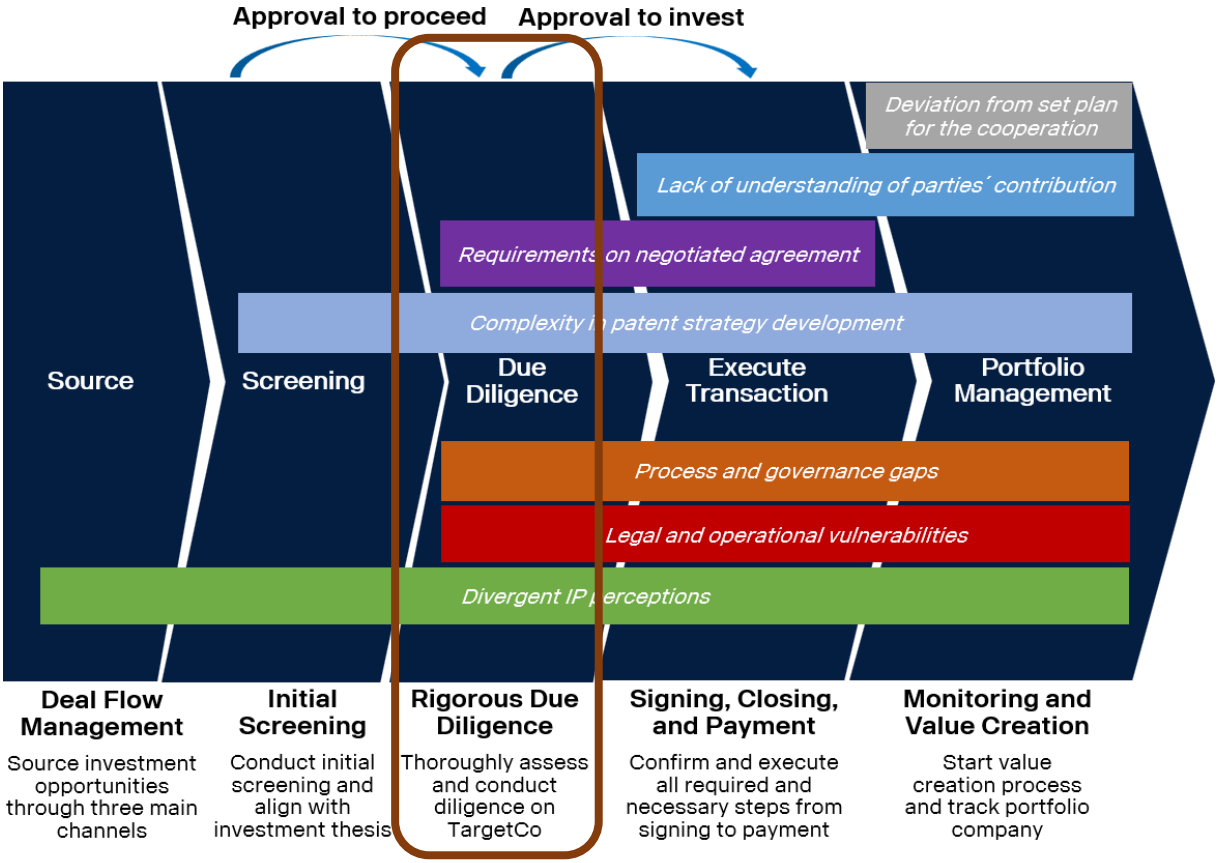
Initial deal terms reflections – Are the deal terms favorable? Do Scania get what is important to them? Should Scania negotiate?

Scania value-add and strategic return – What are the strategic returns Scania can expect, and how do Scania facilitate it?

Exit strategy and financial return – What is the exit plan, and what financial return can Scania expect through that?



3. Due Diligence



Main mission:
Conduct a thorough due diligence that covers all areas to mitigate risks and fully understand TargetCo.

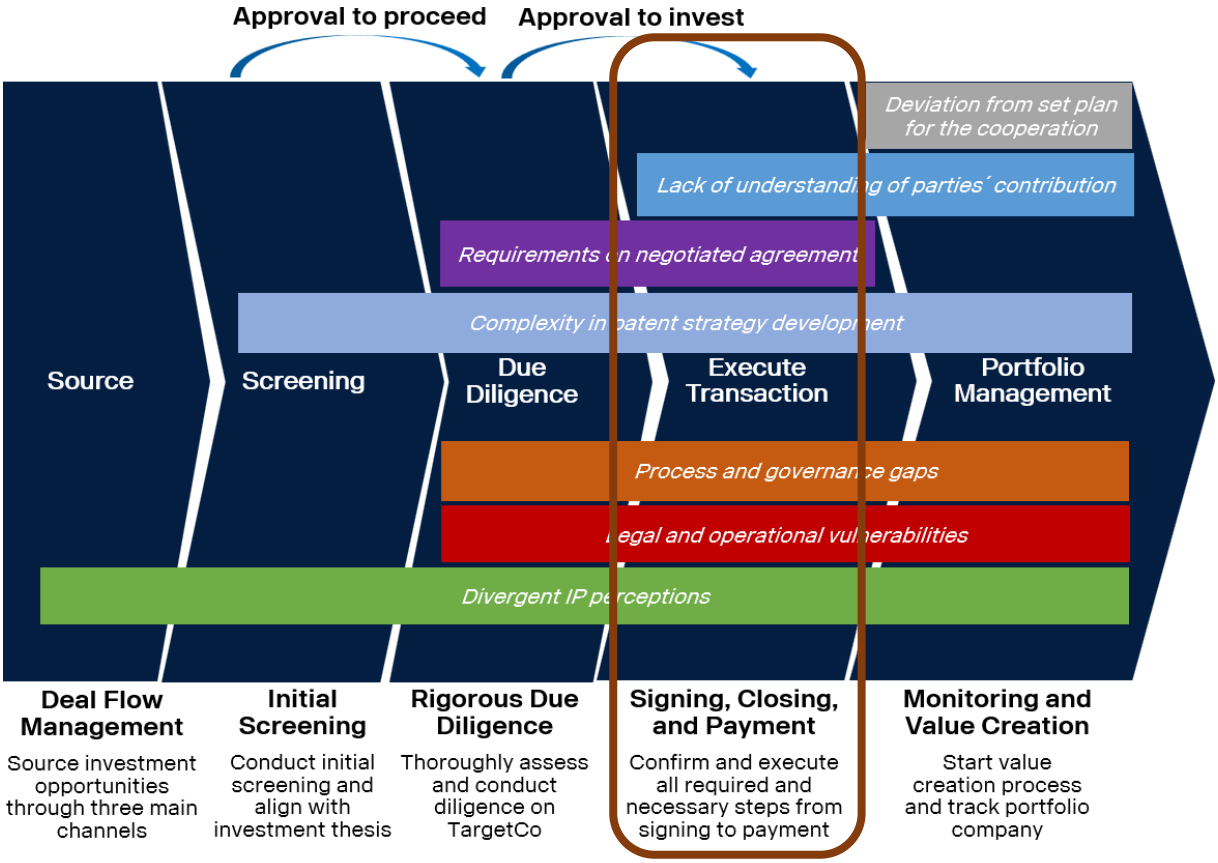
General IP challenges:
IP Strength, IP Ownership, and *IP Leakage.*

Key IP challenges:
Requirements on negotiated agreement, Complexity in patent strategy development, Process and governance gaps, Legal and operational vulnerabilities, and *Divergent IP perceptions.*

Key checkpoints:

- Cross-check data room to *Due Diligence (DD) request list:* Send out *Information Request List (IRL) and Questions and Answers (Q&A)* for missing material
- Invite Group Legal/Legal Team to data room to perform *legal DD*
- Conduct target valuation and *financial DD*
- Invite tech-expert to data room and conduct *tech DD*
- Conduct *commercial* and *strategic DD* and finalise report
- Milestone 1 Volkswagen (VW) reporting

4. Execute Transaction



Main mission:
Execute all necessary and required steps: Signing, closing, and payment.

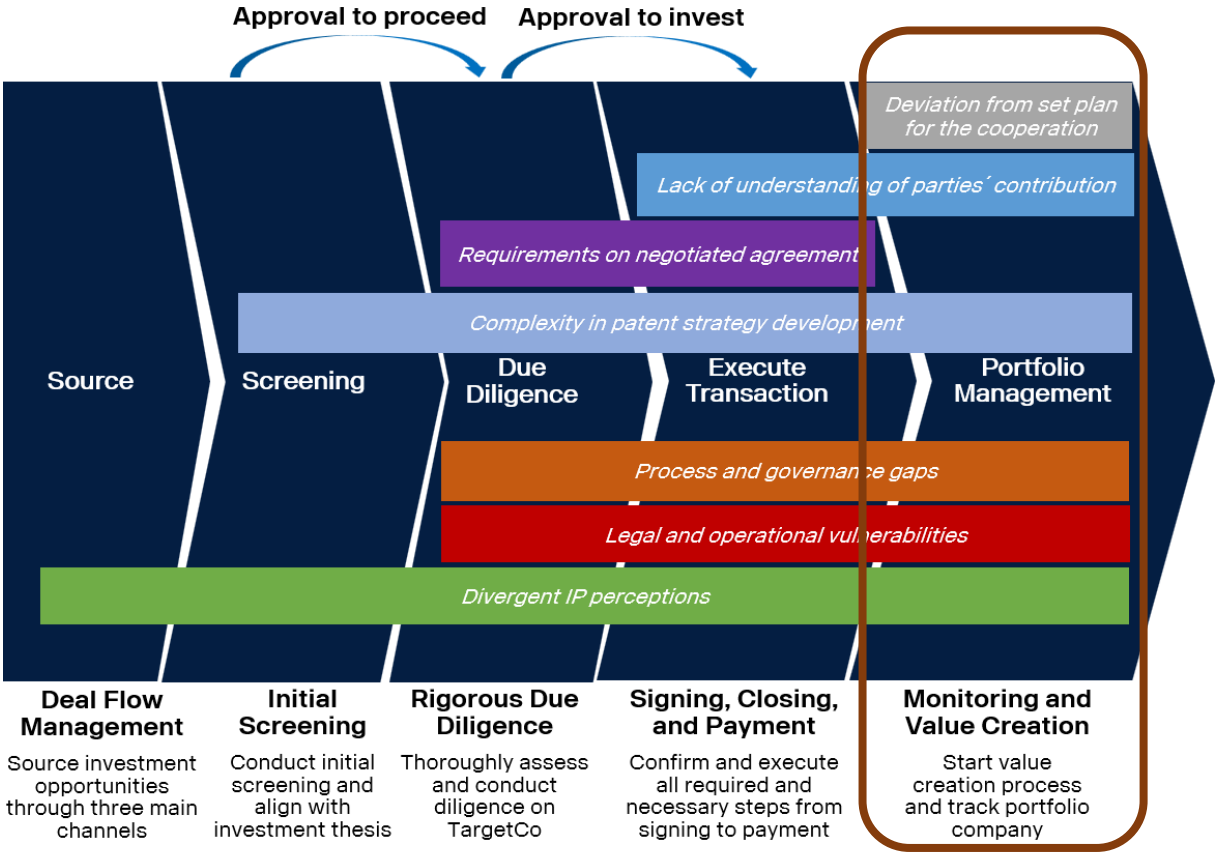
General IP challenges:
IP Ownership, IP Rights Assignment, IP Licensing, and *IP Leakage*.

Key IP challenges: *Lack of understanding of parties´ contribution, Requirements on negotiated agreement, Complexity in patent strategy development, Process and governance gaps, Legal and operational vulnerabilities,* and *Divergent IP perceptions.*

Key checkpoints:

- Confirm *signing* schedule
- Compile DD report and accompanying docs to Approval Package to IC
- Secure "Approval to Invest" from IC
- Notify relevant stakeholders of approval
- Contact Group Treasury for transaction
- Confirm *payment* details with TargetCo and TargetCo bank
- Send *payment* package to Accounts Payable

5. Portfolio Management



Main mission:
Provide value to the portfolio companies across all lifecycle stages, *assess their true value*, *and monitor financial and strategic return continuously*.

General IP challenge:
IP Rights Assignment and *IP Licensing.*

Key IP challenges:
Deviation from set plan for the cooperation, Lack of understanding of parties´ contribution, Complexity in patent strategy development, Process and governance gaps, Legal and operational vulnerabilities, and *Divergent IP perceptions.*

Key checkpoints:
- Identify strategic, operational, and financial levers supporting TargetCo´s value driver
- Assess future value of TargetCo based on successful support of the value levers
- Follow-up commercial milestones and measure financial performance of TargetCo

In *Portfolio Management,* post-investment value creation, as well as tracking and monitoring, are the main focuses. Tracking and monitoring covers *The commercial milestones and traction* and *Financial performance*. These are described below:

The commercial milestones and traction – To understand how the portfolio company is doing in terms of their go-to-market strategy, customer acquisitions, innovative technological breakthroughs – **this helps in building the storyline and creating value.**

Financial performance – To ensure that the portfolio company stays healthy, as well as being on the right **trajectory towards the desired financial return**. Also helps us understand when the next financing round is, **and what support to provide there.**

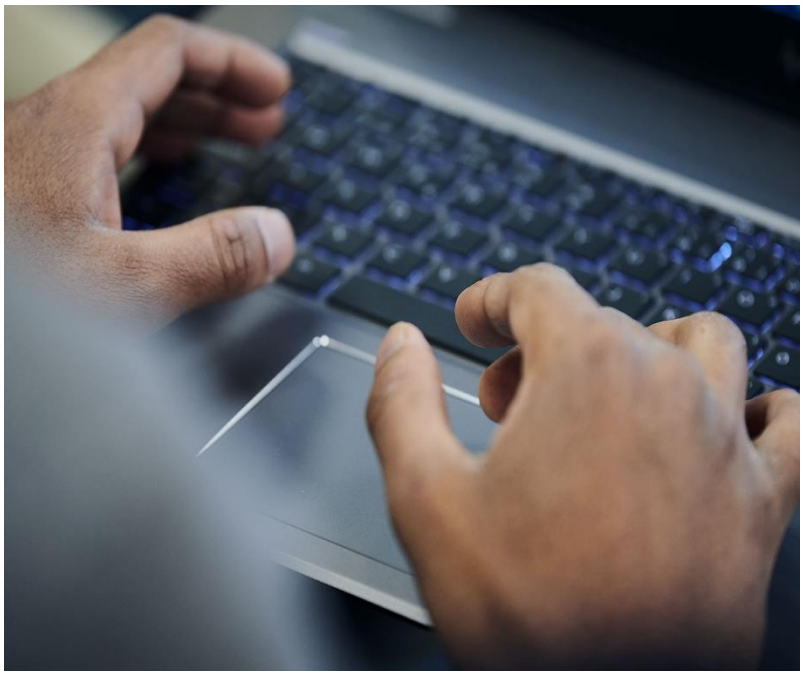
II. Intellectual Property Concepts and Practical Guidelines

1. Methods for Assessment of TargetCo

a) Proof of Concept



- *What is a Proof of Concept (PoC)?*
    - Small-scale project used by TargetCo
- *Benefits of it:*
    - Allows TargetCo to demonstrate the feasibility of their technology or idea for Scania.
    - Helps Scania assess innovation level and technical viability of TargetCo as well as strategic relevance, enabling decision on whether or not to invest.
    - Enables Scania to assess TargetCo´s potential on the market, by identifying potential IP and technical challenges.



- *Preparations and considerations recommended:*
    - Sign NDA before initiating PoC to ensure protection of confidential information shared.
    - Keep communication open and transparent, for instance, by having regular meetings to stay updated.

<u>If there is source code involved:</u>

- Apply encryption techniques to prevent unauthorised access, safeguard IP, and protect code developed during the PoC project.
- Apply access controls to prevent duplication of IP by ensuring authorised individuals are the only ones with permission to view and modify code.

- *Questions to answer:*
    - Does TargetCo have a prototype that properly works (testing of a prototype)?
    - Are there any third-party validations, meaning that other incumbent firms could confirm feasibility and viability?
    - Have TargetCo performed a risk assessment related to commercialisation?

b) Intellectual Property Landscape

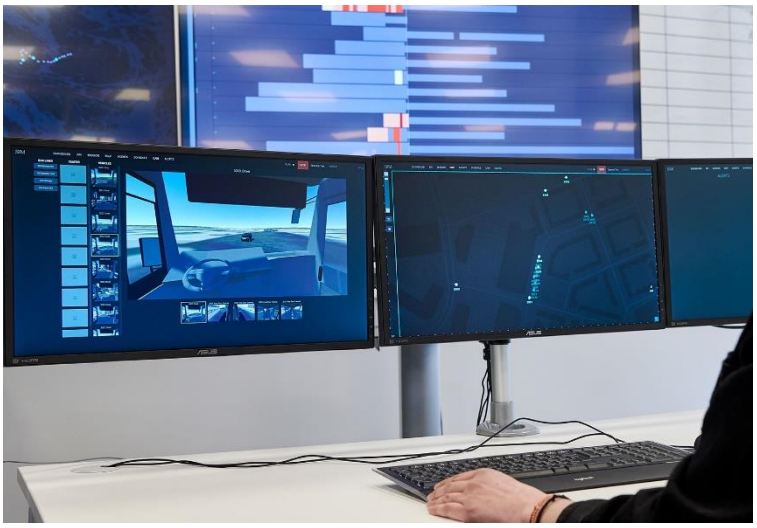
- *What is an IP Landscape?*
    - Comprehensive analysis and evaluation of the IP terrain in a certain tech sector, industry, or market.
    - A systematic search and review of patents, patent applications, scientific literature, and market reports to identify and assess existing IP rights and their impact on business or research activities.



- *Benefits of it:*
    - Identify areas where there is potential competition.
    - Obtain an understanding of the IP environment.
    - Enables strategic decision-making.
- *Questions to answer:*
    - What do TargetCo´s patents, patent applications, and market reports tell about its IP valuation?

c) Freedom to Operate Analysis

- *What is a Freedom To Operate (FTO) Analysis?*
    - An investigation of the possibilities for a product or service to be introduced on the market (market assessment) without the risk of being hampered by others´ rights. FTO is about being able to use a product or process without infringing on existing IP rights.
- *Benefits of it:*
    - Infringement lawsuits could be avoided.
    - Protection of own IP could be improved by understanding other´s IP.
    - Enables successful launch of product.
    - A qualitative FTO analysis could increase the valuation of TargetCo by increasing investor confidence.



- *Questions to answer:*
    - Are there competitors with IP that overlap TargetCo´s IP? Do TargetCo have a competitive advantage?
    - Is it possible for TargetCo to modify its IP (circumvent IP) to avoid legal issues?

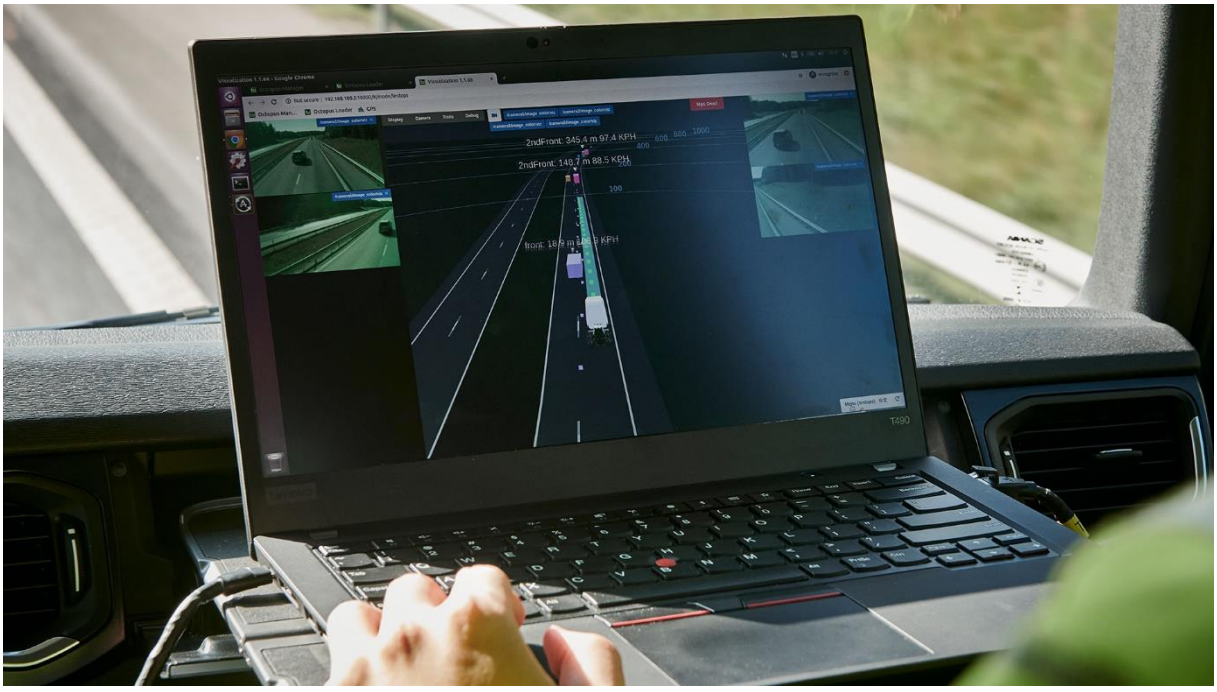
d) Know-How Assessment

- *What is a Know-How Assessment?*
  - Evaluation of the technical expertise of TargetCo, TargetCo´s ability to execute its business plan (business model), and TargetCo´s IP.



- *Benefits of it:*
  - Scania could obtain an understanding of the management team within TargetCo.
  - Scania could evaluate future potential, for instance, if the TargetCo-team has the skills necessary to make TargetCo successful.

- *Steps for this assessment:*
  - Investigate market knowledge, technical skills, and experience of TargetCo´s team.
  - Evaluate TargetCo-team´s capability of making strategic decisions and managing resources.
  - Study TargetCo-team´s ability to build relationships, communicate, and cooperate with investors.



2. Intellectual Property Types and Cooperation Form

a) Background, Foreground, Sideground, and Postground Knowledge

The key IP challenges *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Complexity in patent strategy development*, *Process and governance gaps*, *Legal and operational vulnerabilities*, and *Divergent IP perceptions* are related to the conceptualisation of four IP types. To understand the key IP challenges, it is necessary to understand these: *Background Knowledge*, *Sideground Knowledge*, *Foreground Knowledge*, and *Postground Knowledge*. They are relevant in Research and Development (R&D) collaborations, and are explained below:

- *Background Knowledge* is IP supplied by Scania and TargetCo at the start of *Portfolio Management*.
- *Foreground Knowledge* is IP produced in collaborative research activities, e.g. within TargetCo´s R&D or in a project, during *Portfolio Management*.
- *Sideground Knowledge* is IP produced by any of the partners outside collaborative research activities during *Portfolio Management*.
- *Postground Knowledge* is IP produced by any of the partners after *Portfolio Management*.

b) Alternative Cooperation Forms for Scania and TargetCo

There are several ways to combine the IP types to define the cooperation form for Scania and TargetCo. Some alternatives are given below. These are referred to as "IP structures". The first alternative is presented in *Figure 2: IP structure for collaborative research activities within TargetCo´s business*.

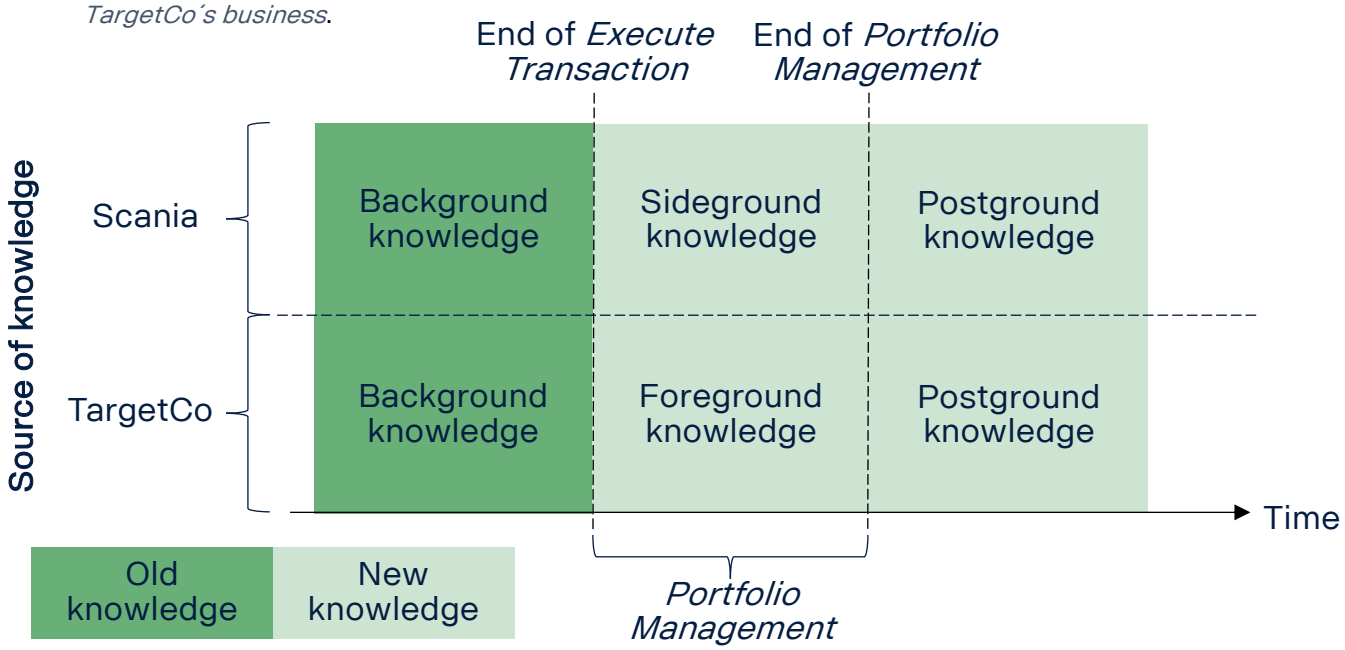


Figure 2: IP structure for collaborative research activities within TargetCo´s business

Here, both Scania and TargetCo provide *Background Knowledge,* which is determined during *Deal Structuring* and used from the start of *Portfolio Management*. The *Foreground Knowledge* is defined as the IP arising from TargetCo´s R&D during *Portfolio Management*. Moreover, there is no *Sideground Knowledge* for TargetCo. This is because the collaborative research activities are performed in TargetCo´s business. Scania Ventures and New Business (VNB) invest capital, resources, and strategic management in TargetCo and its development work, meaning TargetCo is highly dependent on Scania. While maintaining resources for TargetCo, Scania can develop *Sideground Knowledge* useful for the cooperation. After *Portfolio Management, Postground Knowledge* may arise from both Scania and TargetCo.

The second alternative is given in *Figure 3: IP structure that offers simplicity in cooperation*.

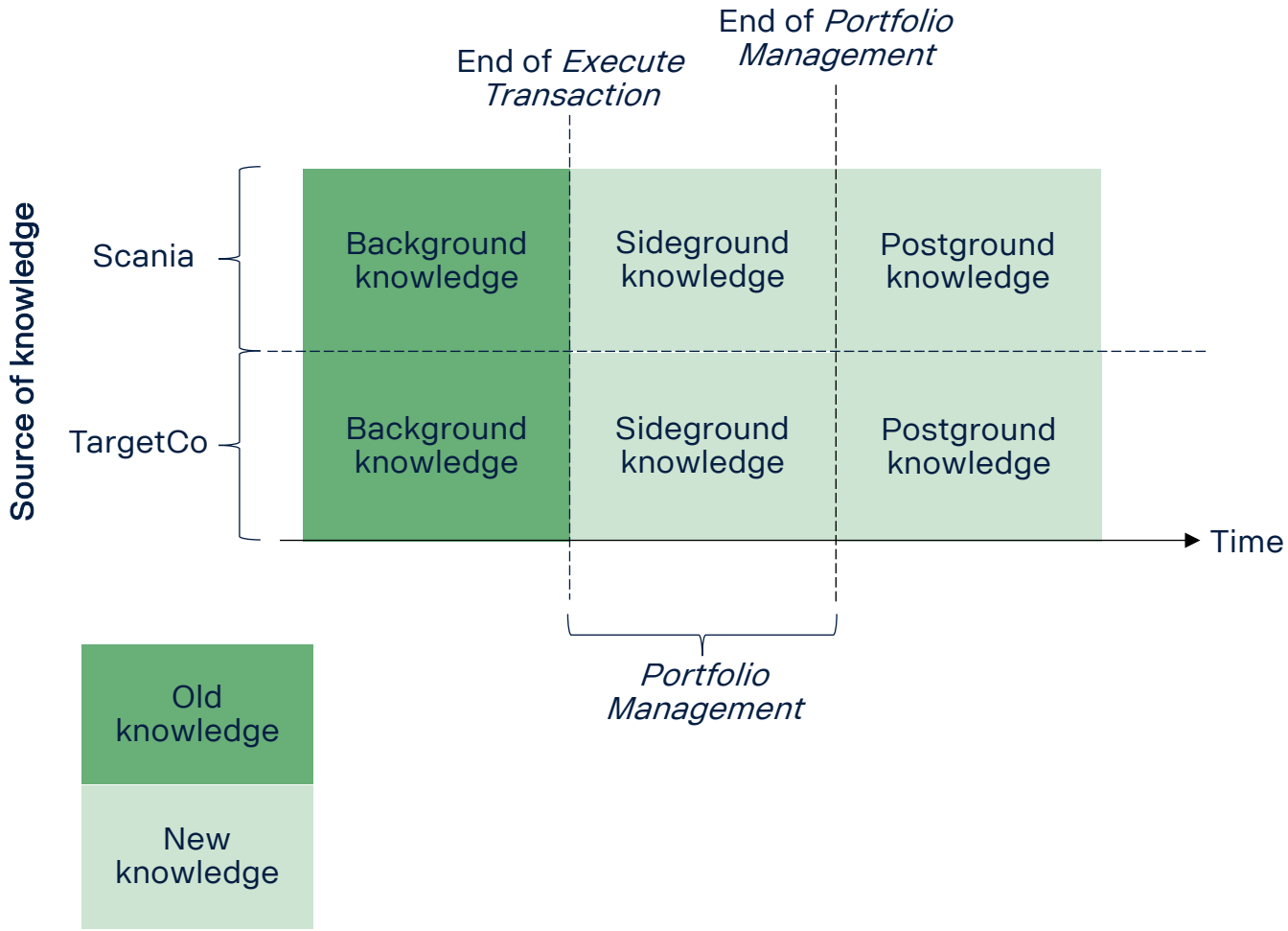


Figure 3: IP structure that offers simplicity in cooperation

Here, there is no *Foreground Knowledge* within the cooperation. Throughout *Portfolio Management*, there is only *Sideground Knowledge* that each party contributes to the cooperation. The expected outcome then consists of dialogues between the parties, where new perspectives, insights, and ideas useful for further development are shared.

The third alternative is given in *Figure 4: IP structure suitable for cooperation in the form of a project*.

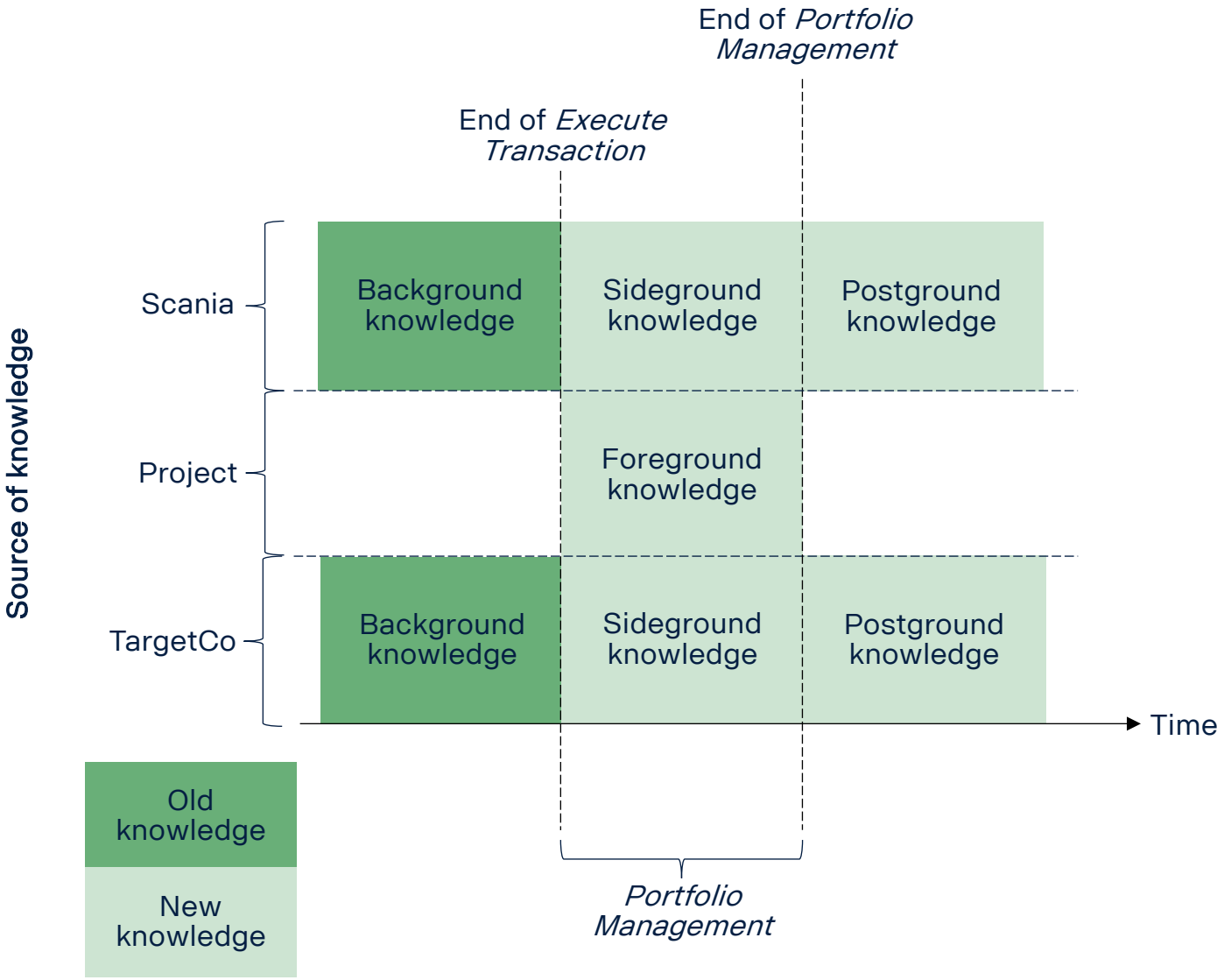


Figure 4: IP structure suitable for cooperation in the form of a project

Here, there is both *Sideground Knowledge* from outside the project and *Foreground Knowledge* within the project throughout *Portfolio Management*. In a project of this kind, Scania and TargetCo combine their knowledge and technologies to develop something new. This form of cooperation could, for instance, be regulated with a JDA. One benefit of choosing this cooperation form is that it enables both parties to cooperate but still conduct R&D on their own (*Sideground Knowledge* for both).

It is also possible to have a combination of cooperation forms. One is presented in *Figure 5: Combination of IP structures, where cooperation in the form of a project is avoided*.

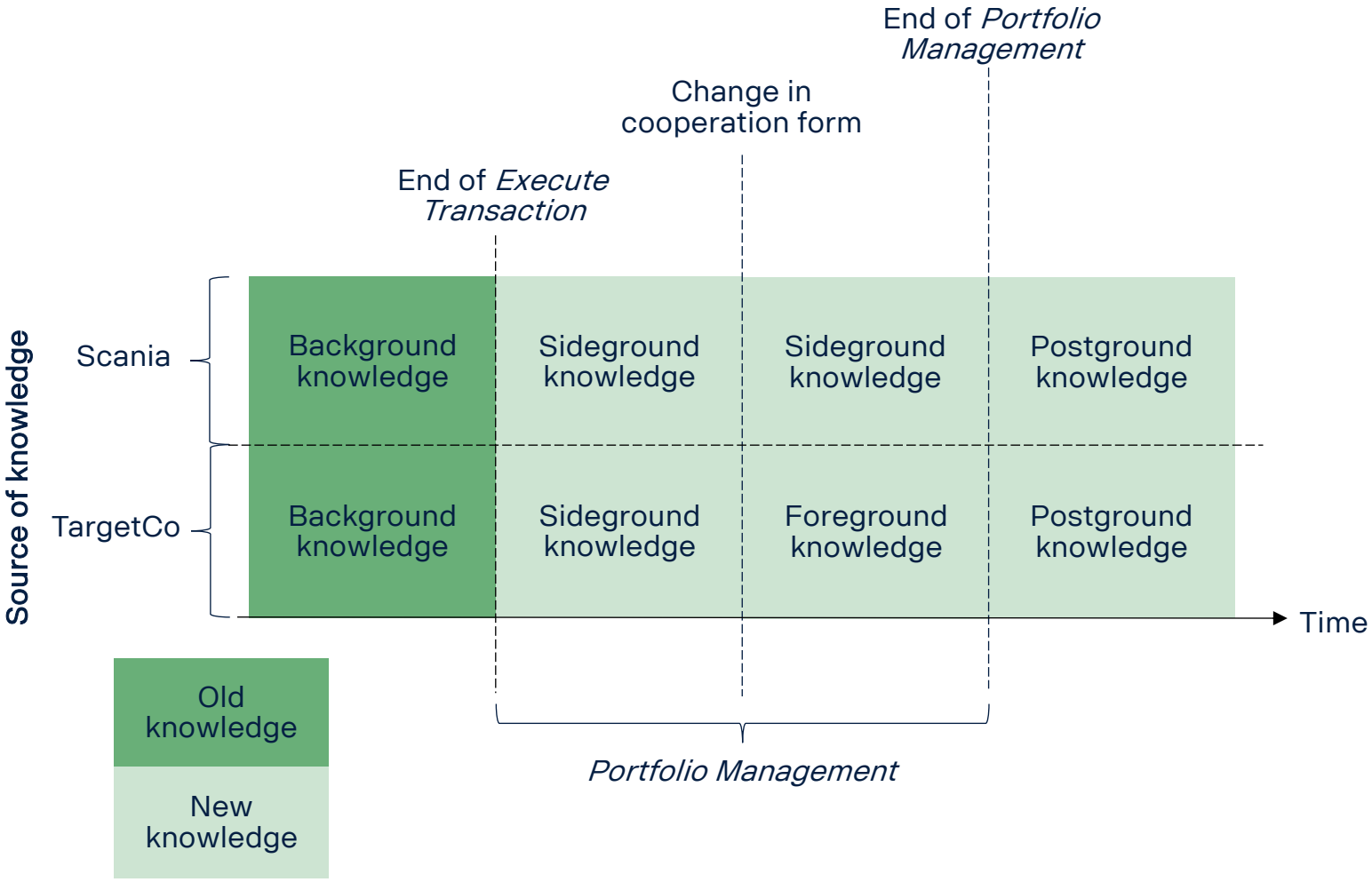


Figure 5: Combination of IP structures, where cooperation in the form of a project is avoided

In this combination, there is only *Sideground Knowledge* at the beginning of *Portfolio Management* (see *Figure 3: IP structure that offers simplicity in cooperation*). After this, there is a change in the cooperation form where *Foreground Knowledge* arises from TargetCo´s R&D instead of *Sideground Knowledge* (see *Figure 2: IP structure for collaborative research activities within TargetCo´s business*). This type of cooperation can be of interest for Scania if they want to initiate cooperation with TargetCo carefully. This means that the cooperation starts lightly with dialogues between each other, where new ideas are being shared, and later continues with more advanced cooperation. Here, Scania supports TargetCo´s R&D with resources and strategic management for the development of *Foreground Knowledge*.

Another combination of cooperation forms is presented in *Figure 6: Combination of IP structures, allowing Sideground Knowledge for both parties*.

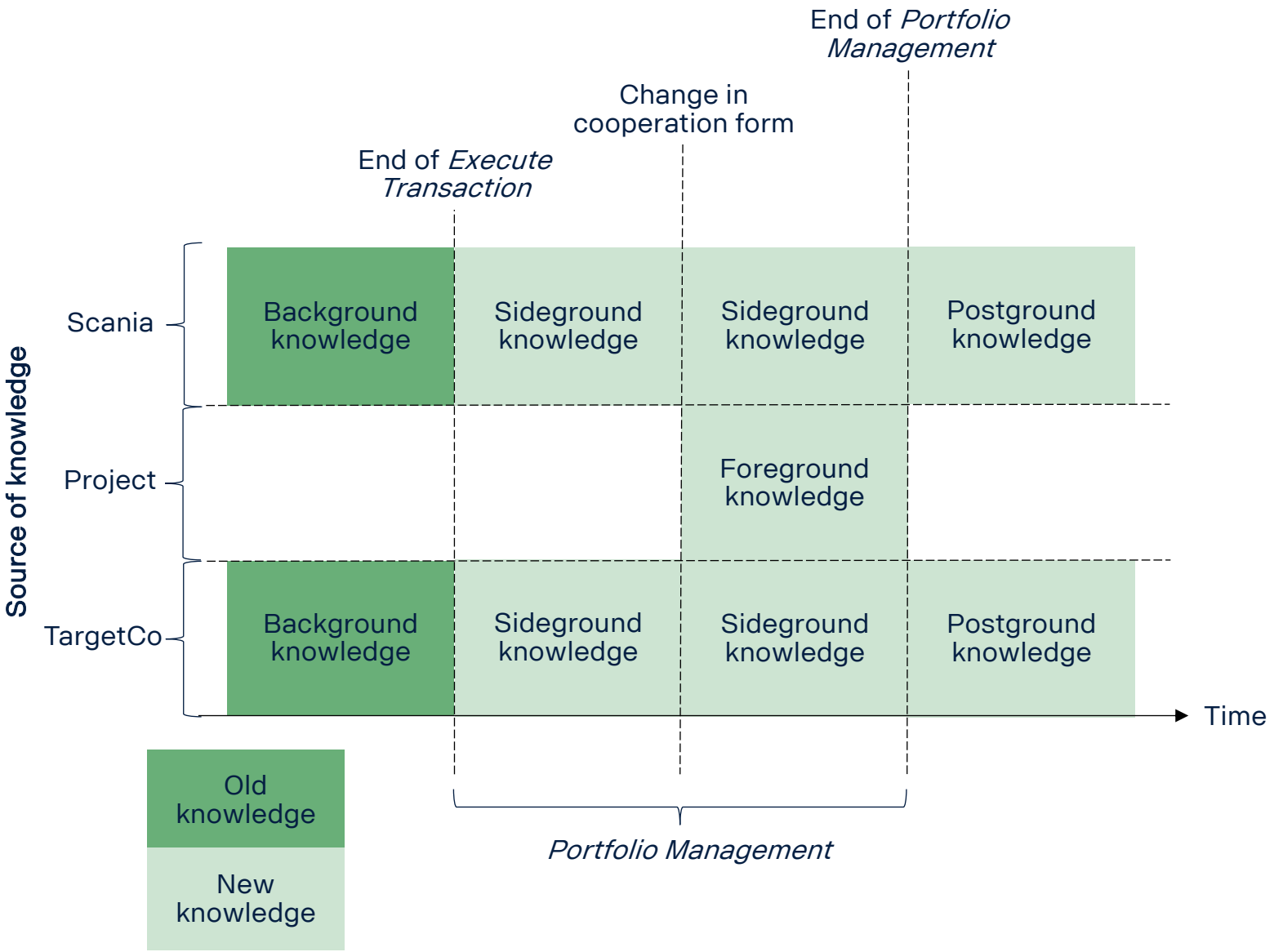


Figure 6: Combination of IP structures, allowing Sideground Knowledge for both parties

In this combination, *Portfolio Management* starts with only *Sideground Knowledge* (see *Figure 3: IP structure that offers simplicity in cooperation*), and then there is a change in the cooperation form where the involved parties enter a project (see *Figure 4: IP structure suitable for cooperation in the form of a project*). This change in cooperation form means that there is both *Sideground Knowledge* and *Foreground Knowledge*. The benefit of applying this cooperation form is that it allows both parties throughout the entire *Portfolio Management* to contribute with *Sideground Knowledge* (i.e. having the opportunity to perform R&D on their own).

Figure 7: Combination of IP structures, where there is a focus on Foreground Knowledge shows a third combination of cooperation forms.

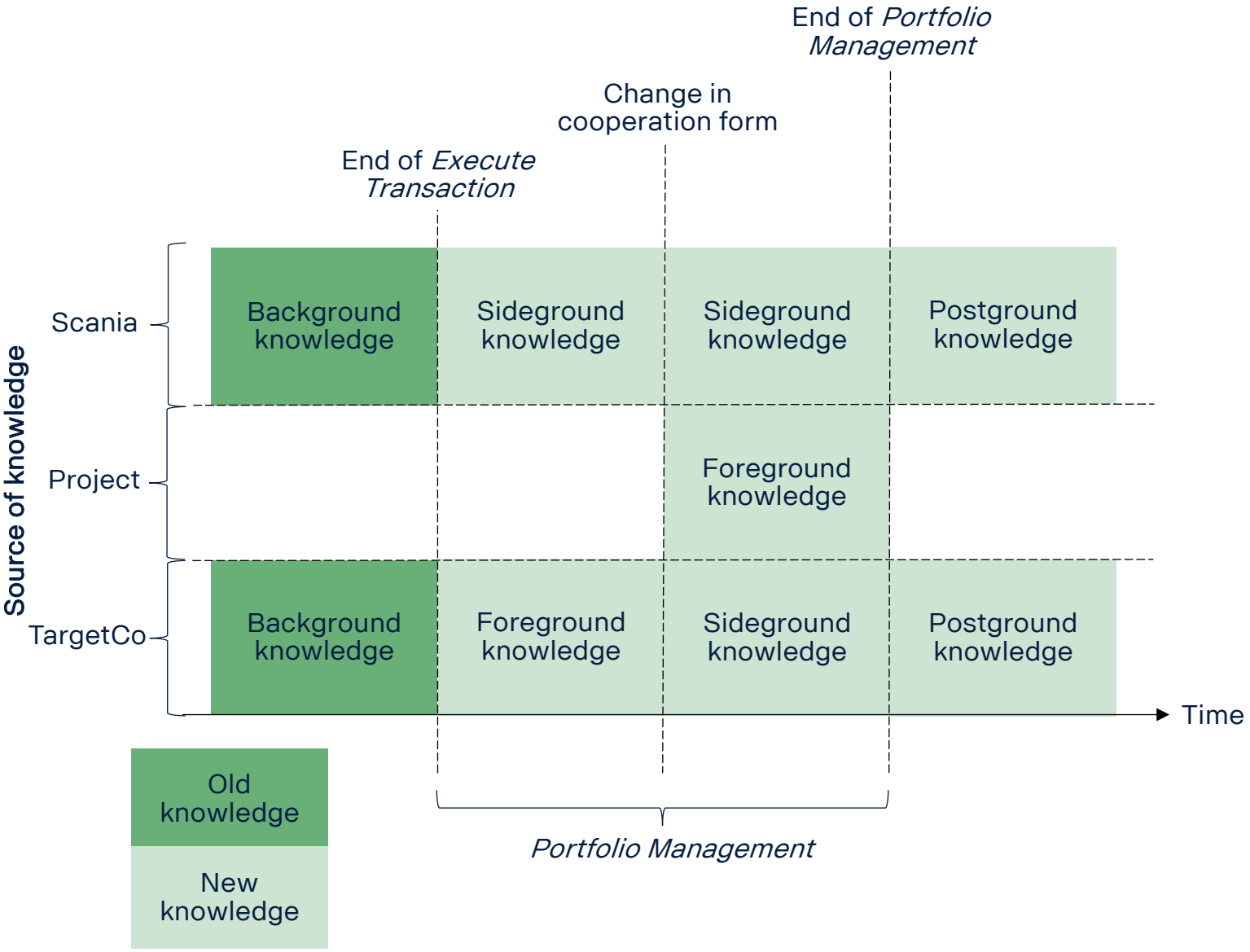


Figure 7: Combination of IP structures, where there is a focus on Foreground Knowledge

In this combination, *Portfolio Management* is initiated with *Sideground Knowledge* from Scania and *Foreground Knowledge* within the R&D of TargetCo (see *Figure 2: IP structure for collaborative research activities within TargetCo´s business*). After this, there is a change in the cooperation form. This change means that the cooperation goes in a different direction, instead of *Foreground Knowledge* at TargetCo´s R&D there are such activities in a project (see *Figure 4: IP structure suitable for cooperation in the form of a project*). Furthermore, a consequence of this transformation is that TargetCo starts to contribute with *Sideground Knowledge* to the cooperation. A reason for making this change could be that TargetCo wants the opportunity to perform R&D on their own (*Sideground Knowledge*).

Important to note is that in addition to the combinations of cooperation forms presented in *Figure 5: Combination of IP structures, where cooperation in the form of a project is avoided*, *Figure 6: Combination of IP structures, allowing Sideground Knowledge for both parties*, and *Figure 7: Combination of IP structures, where there is a focus on Foreground Knowledge*, there are other alternative combinations. For instance, it is possible to switch places between the two IP structures inside *Portfolio Management*. Another way of changing the cooperation form is by adding new parts to *Portfolio Management*. This can be done either by adding an IP structure that has already been used or by adding a new one. Potential reasons for doing this:

- The parties of the cooperation form want to return to an IP structure that has been used before, because they realise that one is better
- The parties want to try a new cooperation form, to see if that one is more aligned with the visions and goals of the cooperation.



3. General Intellectual Property Challenges

Based on the IP structures, several key IP challenges were identified. They need to be considered and managed carefully for efficient cooperation between Scania and TargetCo. They are: *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Complexity in patent strategy development*, *Process and governance gaps*, *Legal and operational vulnerabilities*, and *Divergent IP perceptions*. Before explaining these, the general IP challenges *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage* are explained:

a) Intellectual Property Strength

IP Strength is about evaluating TargetCo.
More specifically:



1. <u>What to do</u>:
   Evaluate TargetCo´s routines, patent portfolio, and know-how (i.e. the knowledge and skills needed to implement inventions).



2. <u>How to do it</u>:
   Let TargetCo demonstrate the feasibility of their technology or idea for Scania through a *PoC*.

   Perform an *IP Landscape* before investing in a certain technical field.

   Check if a *FTO* is conducted by TargetCo.

   Conduct a *Know-How Assessment* to evaluate TargetCo.



3. <u>When to do it</u>:
   Source, Screening, and *Due Diligence*.



4. <u>Targeted IP:</u>
   Background Knowledge.

b) Intellectual Property Ownership

IP Ownership is about performing research on TargetCo to see that it has a clear title to its IP. It also covers the negotiation of terms regulating the usage of this IP and Scania´s IP. More specifically:



1. <u>What to do</u>:
   Ensure that there are no third-party ownership claims.



2. <u>How to do it</u>:
   Check the status by asking the following questions to TargetCo:

   - Are there any ongoing or potential IP disputes?
   - Are you the sole owner of the IP listed?
   - Can you verify your ownership of all provided IP, including any assignment from any creator, author, and/or inventor?
   - Is your right to use the provided IP limited by or dependent on any third party?
   - Is any of the provided IP subject to any license agreement or similar?
   - Is the use of any of the provided IP limited by any rights held by third parties?

   Negotiate terms for the usage of Scania´s and TargetCo´s IP.



3. <u>When to do it</u>:
   Screening, *Due Diligence,* and *Deal Structuring (Execute Transaction)*.



4. <u>Targeted IP:</u>
   Background Knowledge.

c) Intellectual Property Rights Assignment

IP Rights Assignment is about the way of regulating the rights assignment of new IP arising within the cooperation between Scania and TargetCo.
More specifically:



1. <u>What to do</u>:
   Regulate the rights assignment of IP that arises within the cooperation.



2. <u>How to do it</u>:
   Develop an agreement with TargetCo that manages the rights assignment:
   - For example, a Joint Development Agreement (JDA). It should clearly state who owns what.
   - Ensure that the agreement is well thought out, since it determines which party has control of licensing, etc.



3. <u>When to do it</u>:
   Deal Structuring (Execute Transaction) and *Portfolio Management*.



4. <u>Targeted IP:</u>
   Foreground Knowledge, *Sideground Knowledge*, and *Postground Knowledge*.

d) Intellectual Property Licensing

IP Licensing and its strategic management are essential.
More specifically:



1. <u>What to do:</u>
   Negotiation of license agreements which satisfy Scania and TargetCo.



2. <u>How to do it:</u>
   Ask the following questions when defining the license scope:

   - What degrees of freedom does TargetCo need for them to feel that the license scope is aligned with their long-term business objectives and market potential?
   - Do TargetCo expect scalable license agreements to have expansive opportunities?
   - What expectations do Scania have of the license scope?

   Use, for instance, *Change-of-Technology Clauses* in the agreement to secure usage rights:

   - *Assign-Back Clause* stipulates that ownership of improvements of the licensed technology made by the licensee must be transferred to the licensor.
   - *Grant-Back Clause* stipulates that improvements of the licensed technology made by the licensee must be licensed to the licensor.
   - *Grant-Forward Clause* stipulates that the licensee must be offered a license on improvements of the licensed technology made by the licensor.

   To outline how the parties are allowed to use different IP, a *Change-Of-Control Clause* would be useful:

   - A clause of this form could stipulate that the licensor has the right to terminate the license agreement in case of a change of ownership of the licensee.



3. <u>When to do it:</u>
   Deal Structuring (Execute Transaction) and *Portfolio Management*.



4. <u>Targeted IP:</u>
   Background Knowledge, *Foreground Knowledge*, *Sideground Knowledge*, and *Postground Knowledge*

e) Intellectual Property Leakage

IP Leakage needs to be avoided since
it refers to the leaking of sensitive information
to unauthorised parties, which is a threat to the cooperation.
More specifically:



1. <u>What to do</u>:
   Ensure that all parties within the cooperation protect sensitive business and technical information.



2. <u>How to do it</u>:
   Ask the following questions:

   - What measures has TargetCo taken to protect core IP assets (i.e. management of patent applications and safeguarding of trade secrets)?
   - Does TargetCo store, monitor, and restrict access to source code as well as other valuable data?

   Prepare and sign a Non-Disclosure Agreement (NDA):

   - Does the NDA cover all aspects? Will it protect sensitive business and technical information?



3. <u>When to do it</u>:
   Screening, Due Diligence, and *Deal Structuring (Execute Transaction)*.



4. <u>Targeted IP:</u>
   Background Knowledge, *Foreground Knowledge*, *Sideground Knowledge,* and *Postground Knowledge*.

4. Key Intellectual Property Challenges

With the concepts of IP structures, *IP Strength*, *IP Ownership*, *IP Rights Assignment*, *IP Licensing*, and *IP Leakage explained*, the key IP challenges *Deviation from set plan for the cooperation*, *Lack of understanding of parties´ contribution*, *Requirements on negotiated agreement*, *Complexity in patent strategy development*, *Process and governance gaps*, *Legal and operational vulnerabilities*, and *Divergent IP perceptions* are explained below:

a) Deviation From Set Plan for the Cooperation



1. **Key IP challenge:**
   Deviation from set plan for the cooperation.

2. **Description of challenge:**
   Changes need to be made to the set plan for the cooperation.

3. **Steps of VC investment process where challenge is managed:**
   Deal Structuring (Execute Transaction) and *Portfolio Management.*

4. **Reasons for challenge to occur:**
   The cooperation form that the parties chose for the cooperation no longer meets their needs.

5. **Effects of challenge occurring:**
   Change in cooperation form involves several steps and could therefore be quite complicated to execute. With contingency plans prepared, however, it is possible to execute this smoothly. A change in cooperation form affects the knowledge produced within the cooperation (*Foreground Knowledge* and *Sideground Knowledge*). Moreover, this impacts *IP Rights Assignment, IP Licensing,* and *IP Leakage* because the IP arising during the cooperation now differs from the previous plan. More specifically, this difference in IP type impacts the criteria for the assignment, licensing, and protection of the IP.

6. **Management of challenge:**
   Management during *Deal Structuring:*

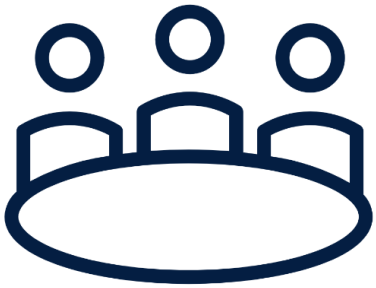
   - Jointly develop contingency plans for managing changes in the cooperation form. For instance, one part of the plan could be suggestions on cooperation forms (IP structures) to change to.

   Management during *Portfolio Management:*

   - Execute the contingency plans prepared and included in the contract.

b) Lack of Understanding of Parties´ Contribution

1. Key IP challenge:
   Lack of understanding of parties´ contribution.



2. Description of challenge:
   There is a risk that a lack of understanding of the contributions of each party arise.

3. Steps of VC investment process where challenge is managed:
   Deal Structuring (Execute Transaction) and *Portfolio Management.*

4. Reasons for challenge to occur:
   The parties have different intentions with the cooperation. For example, one party´s intention with the cooperation could be to learn from each other to develop IP on their own *(Sideground Knowledge).* Meanwhile, the other party might expect the cooperation to have a project *(Foreground Knowledge)* in addition to this. It can also be a lack of understanding related to contribution in terms of capital, management, or R&D resources.

5. Effects of challenge occurring:
   This lack of understanding could lead to conflicts between the parties, which hamper the cooperation. The lack of understanding could, for instance, be related to the parties' contribution to the *Foreground Knowledge* produced, affecting *IP Rights Assignment*, or *IP Licensing* of *Background Knowledge*.

6. Management of challenge:
   Management during *Deal Structuring:*

   - Discuss questions related to contribution expectancy. Below, three questions to discuss are suggested:

     1. In terms of capital, management, and R&D resources, what is Scania expected to contribute?
     2. When TargetCo wants to raise capital, for instance, through a new share issue, should there be any milestones that TargetCo must achieve? If yes, what are the milestones that have to be achieved for further financing?
     3. What *Background Knowledge* is each party expected to bring into the cooperation? How should the framework for *Sideground Knowledge*, *Foreground Knowledge*, and *Postground Knowledge* be designed?

   Management during *Portfolio Management:*

   - There is a risk that a lack of understanding of the parties´ contribution arises during the cooperation. To manage this, communication and willingness to compromise are key.

c) Requirements on Negotiated Agreement



1. Key IP challenge:
   Requirements on negotiated agreement.

2. Description of challenge:
   Depending on the complexity of the cooperation form, the requirements for the negotiated agreements vary.

3. Steps of VC investment process where challenge is managed:
   Due Diligence and *Deal Structuring (Execute Transaction)*.

4. Reasons for challenge to occur:
   A cooperation form (IP structure) which contains several variants of *Foreground Knowledge* (i.e. innovative work made together within a project or TargetCo´s R&D) increases the requirements on the agreement because this type of knowledge involves much management.

5. Effects of challenge occurring:
   During *Deal Structuring*, many requirements on the agreement mean there are several aspects to cover. Moreover, it is important that the agreement cover management of *IP Ownership, IP Rights Assignment, IP Licensing,* and *IP Leakage*. Depending on IP structure, i.e. *Background Knowledge, Foreground Knowledge, Sideground Knowledge,* and *Postground Knowledge*, the requirements on this vary.

6. Management of challenge:
   Management during *Due Diligence:*

   - Note characteristics of TargetCo that potentially impact the requirements of an agreement. These characteristics could, for instance, impact Scania´s expectations on the cooperation, thereby also impacting the requirements of the agreement.

   Management during *Deal Structuring:*

   - Apply clauses to manage requirements on the agreement. To manage these requirements is necessary for the agreement to be aligned with the strategic goals of the cooperation.

d) Complexity in Patent Strategy Development



1. Key IP challenge:
   Complexity in patent strategy development.

2. Description of challenge:
   There are several challenges involved in the development of a patent strategy.

3. Steps of VC investment process where challenge is managed:
   Screening, Due Diligence, Deal Structuring (Execute Transaction), and *Portfolio Management.*

4. Reasons for challenge to occur:
   Depending on whether there is *Sideground Knowledge* or *Foreground Knowledge*, if the *Foreground Knowledge* arises from a project or within TargetCo´s R&D, etc., the complexity of the patent strategy varies. Moreover, different opinions about patent coverability, i.e. which markets to obtain IP protection in and what technology to protect, as well as which way to protect the IP make the patent strategy development quite challenging.

5. Effects of challenge occurring:
   The patent strategy development is critical and is therefore crucial for the outcome of the cooperation. Furthermore, there is a risk that a conflict of interest arises between Scania and TargetCo. It could be that Scania expects it to be easy for competitors to circumvent patent protection or that it is hard to prove patent infringement, and therefore suggests that the *Foreground Knowledge* should be kept a trade secret. This might not align with TargetCo's interest, which instead wants to secure a patent on the *Foreground Knowledge* before licensing it to other parties outside the cooperation to obtain revenue streams. TargetCo might be highly dependent on revenue streams from the *Foreground Knowledge* due to its small size and unestablished business, meanwhile, Scania might prefer to avoid licensing to competitors. Here, *IP Rights Assignment* and *IP Licensing* are affected.

6. Management of challenge:
   Management during *Screening:*

   - Create an idea of TargetCo´s patent strategy approach through a know-how assessment. Studying the IP of TargetCo could give a hint of their expectations of Scania in the joint development of a patent strategy. Interesting to know is whether TargetCo has trade secrets or not, typical geographic scope, and patent coverage.

   Management during *Due Diligence:*

   - Same investigation as in *Screening*, however, on a more detailed level.

   Management during *Deal Structuring:*

   - Jointly develop an agreement which enables patent strategy development that satisfies the goals and visions of the cooperation.

   Management during *Portfolio Management:*

   - Discuss the approach for disclosing and protecting IP produced during the cooperation. Ensure alignment on how and when inventions are disclosed to avoid anticipating patent applications, as well as how to protect different types of IP (guidelines for when to file for patents and when to keep secret).

e) Process and Governance Gaps



1. <u>Key IP challenge:</u>
   Process and governance gaps.

2. <u>Description of challenge:</u>
   Contractual gaps as well as cultural and operational misalignment are risks involved.

3. <u>Steps of VC investment process where challenge is managed:</u>
   Due Diligence, Deal Structuring (Execute Transaction), and *Portfolio Management.*

4. <u>Reasons for challenge to occur:</u>
   Delays in formalising agreements, unclear *IP Rights Assignment* within cooperation, and a lack of responsibility in annual fee payment in case joint ownership of IP is applied are all reasons for having contractual gaps.

5. <u>Effects of challenge occurring:</u>
   There is a risk that conflicts arise due to disrespectful behaviour from any of the parties, which could negatively impact the outcome of the cooperation.

6. <u>Management of challenge:</u>
   Management during *Due Diligence:*

   - Consider the risk of TargetCo having non-public patent applications when performing *Due Diligence*. Scania is unable to find non-public patent applications, which hinders *Due Diligence*. This means Scania cannot obtain an understanding of the total patent portfolio.

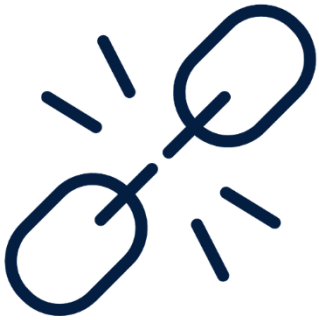
   Management during *Deal Structuring:*

   - Jointly develop a plan which covers IP structures possible to use as representation for the cooperation form to avoid unclear *IP Rights Assignment*.
   - Set up processes for regular monitoring of the agreement and how the parties handle their respective obligations.
   - Consider a clause in the agreement which gives the right to terminate if the other party fails to meet the key obligations.
   - Ensure that the parties agree on the meaning of the terms involved in the agreement, where there is room for interpretation, and why that is the case.

   Management during *Portfolio Management:*

   - Have open and transparent communication, meaning there is a dialogue between the parties to address concerns before they escalate.
   - Take necessary measures to keep the cooperation running smoothly.
   - Management of *Postground Knowledge* is essential, especially when reaching investment exit. Ensure to manage this in time.

f) Legal and Operational Vulnerabilities



1. Key IP challenge:
   Legal and operational vulnerabilities.

2. Description of challenge:
   This challenge involves risks related to patent quality and coverage, IP contamination, as well as open source and licensing.

3. Steps of VC investment process where challenge is managed:
   Due Diligence, Deal Structuring (Execute Transaction), and *Portfolio Management.*

4. Reasons for challenge to occur:
   The use of open source code, third-party involvement, and not considering each party´s interests are some examples of possible reasons.

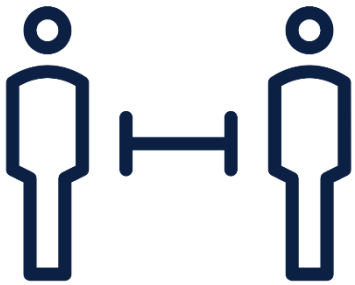
5. Effects of challenge occurring:
   - There is a risk that a patent protects what one party wants to do, but not the other party´s interests.
   - IP contamination is the involuntary transfer of IP between parties. This causes problems within the cooperation and should be avoided.
   - Open source software has transformed how technology is developed and used. It facilitates cooperation that allows developers to access, modify, and distribute code. Quite often, there is no need to pay licence fees to an upstream licensor. The open source model could, however, clash with the traditional IP practices that aim to secure a monopoly for innovators. This could be problematic and should be considered.
   - There is a risk that TargetCo use Scania´s brand without authorisation, which could negatively impact Scania´s reputation.

6. Management of challenge:
   Management during *Due Diligence:*
   - Investigate TargetCo´s use of open source software. If open source software is used, IP risks could be exposed.
   - Investigate if the coverage of TargetCo´s IP is aligned with the strategic vision of both Scania and TargetCo, meaning that the patent protects what Scania and TargetCo want to do together.

   Management during *Deal Structuring:*
   - Do risk management of IP contamination, open source software, and licensing.
   - Be transparent in the negotiation to avoid accidental infringement and contamination from undisclosed IP.
   - Consider a license structure in the agreement which focuses on field-of-use.

   Management during *Portfolio Management:*
   - Follow the agreement, e.g. the framework for *IP Rights Assignment*.
   - When planning for investment exit, to ensure everyone is on the same track is critical. To avoid TargetCo licensing IP to competitors or using Scania´s brand without authorisation, it is beneficial to create clarity before investment exit and follow up on each party´s work post investment exit.

g) Divergent Intellectual Property Perceptions



1. Key IP challenge:
   Divergent IP perceptions.

2. Description of challenge:
   There may be differences between Scania and TargetCo in IP knowledge, IP valuation, and the attitude towards exclusivity. IP knowledge and IP valuation are related to the level of expertise in IP, estimation of patent scope and its value, and the use of FTO analysis. Exclusivity is directly linked to licensing, where TargetCo may prefer to give out non-exclusive licenses to keep the opportunity to work with others.

3. Steps of VC investment process where challenge is managed:
   Source, Screening, Due Diligence,
   Deal Structuring (Execute Transaction), Portfolio Management.

4. Reasons for challenge to occur:
   The parties have different backgrounds and experiences, causing them to have different perceptions.

5. Effects of challenge occurring:
   Different IP perceptions could cause the parties to avoid cooperation because it is too hard to manage.

6. Management of challenge:
   Management during *Source:*
   - When sourcing the deal flow, it is beneficial to pay attention to how TargetCo understands the value of its IP. This is one step in ensuring that its IP is aligned with Scania´s strategic goals. To obtain an idea of TargetCo´s IP valuation, one could perform an IP landscape as well as an assessment of TargetCo´s patent portfolios.

   Management during *Screening:*
   - Analyse geographic coverage and conduct a know-how assessment. This type of assessment contains an investigation of TargetCo´s IP, team, market capabilities, position compared to competitors, and operational & executional capabilities. When investigating IP, it is important to do FTO and evaluate if TargetCo has FTO analysis routines to avoid IP infringement.

   Management during *Due Diligence:*
   - Same investigation as in *Screening*, however, in more detail.

   Management during *Deal Structuring:*
   - Be willing to make compromises when negotiating to have a chance of solving potential differences in IP perception.

   Management during *Portfolio Management:*
   - Be able to accept changes to keep the cooperation moving forward. For instance, if one of the parties might find an issue with the cooperation form and has an interest in changing it.

   - In the process of preparing an investment exit, there are several agreements which have to be checked and finalised. The management of *Postground Knowledge* is essential, since it is likely that this IP includes improvements made on *Background Knowledge*, *Foreground Knowledge*, and *Sideground Knowledge*.

GLOBAL INTELLECTUAL PROPERTY PLAYBOOK

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